When Do Foundation Models Pay Off for Retail Demand Forecasting?
A Systematic Review and Cross-Benchmark Meta-Analysis

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Abstract

We present a PRISMA-informed systematic review of forecasting methods for short-term retail demand, combined with original gap-filling experiments and cross-benchmark extraction from fev-bench and GIFT-Eval. We screen approximately 150 papers (2020–2026) and extract quantitative results into 800+ rows spanning 9 retail datasets. A model-level meta-regression (`metafor::rma.mv`, REML, Knapp–Hartung) on $k = 543$ paired comparisons across foundation models and conventional baselines reveals a significant FM advantage on both distributional and point-forecast metrics, with a substantially stronger signal on the former: on Scaled Quantile Loss (SQL), $\hat{\Delta} = -0.318$ ($p < .0001$); on WAPE, $\hat{\Delta} = -0.116$ ($p = 0.018$). The full-pool aggregate ($\hat{\Delta} = -0.195$, $p < .0001$) confirms a net FM advantage. Foundation models achieve substantially lower quantile-loss scores; their point forecasts are also significantly better but the advantage is attenuated relative to distributional metrics. Model scale (9 M–200 M parameters) is not a significant moderator in the retail domain. We provide a practitioner-oriented cost–accuracy Pareto analysis on consumer hardware and document that the LightGBM direct-vs-recursive protocol gap is conditional on demand intermittency.

Keywords: demand forecasting, foundation models, meta-analysis, LightGBM, systematic review, retail, time series

1. Introduction

The 2024–2026 wave of time-series foundation models — Ansari et al. (2025), TimesFM 2.5 (Das et al., 2025), Moirai 2.0 (Liu et al., 2025), TiRex (Auer et al., 2025) — has shifted the default assumption in short-term forecasting. Where a retail team in 2022 would start with LightGBM and a feature-engineering sprint, the 2026 playbook increasingly says: try a zero-shot foundation model first, tune LightGBM only if the FM disappoints. Each model paper reports favourable results on its own evaluation suite — GIFT-Eval, fev-bench, or a curated subset of the M competitions — but no paper runs the same battery of models under matched conditions on the same retail datasets and reports the delta. The benchmarks multiply; the synthesis does not.

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This paper asks: **when do foundation models actually pay off for retail demand forecasting?** We operationalise "pay off" as a statistically significant reduction in forecast error relative to a well-tuned baseline (gradient-boosted tree or statistical model) under matched evaluation conditions. By "when" we mean: on which datasets, horizons, demand distributions, metrics, and hardware budgets.

1.1. Research questions

We address four research questions across three complementary lenses (accuracy, moderators, and cost):

- **RQ1 (Accuracy):** Do zero-shot foundation models outperform conventional baselines on retail demand forecasting? (§6.2)
- **RQ2 (Moderators):** Which characteristics — metric choice, demand intermittency, model scale, forecast horizon — moderate the FM-vs-baseline gap? (§6.8)
- **RQ3 (Cost):** What is the cost-accuracy Pareto frontier when consumer-hardware electricity, cloud-GPU list price, and CO_2 footprint are included? (§6.7)
- **RQ4 (Guidance):** Under what conditions should a retail forecasting team deploy a foundation model instead of investing in feature engineering for LightGBM? (§8)

1.2. Approach

This study combines a PRISMA-informed systematic review with original gap-filling experiments and structured extraction from two large forecasting benchmarks (fev-bench, GIFT-Eval).

1. **Source A (literature + benchmark extraction).** We screen ~150 papers (2020–2026) from Scopus, Google Scholar, Semantic Scholar, and arXiv, applying retail-domain inclusion criteria. We supplement the extraction with per-model per-task results from fev-bench (Shchur et al., 2025) (20 retail tasks, 7 FMs vs LightGBM/CatBoost baselines) and GIFT-Eval (Aksu et al., 2024) (4 Sales-domain tasks, 7 FMs vs statistical baselines). The combined extraction yields 800+ rows across 9 retail datasets (§3.2).
2. **Source B (gap-filling experiments).** We run our own experiments under a unified evaluation protocol (§5.1): five foundation models (Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2, TimesFM 2.5; 9–200 M parameters) on a consumer MacBook (Apple Silicon MPS / CPU), and three baselines (lightgbm_cov, lightgbm_direct, seasonal_naive) on Azure ML. Source B produces gap-filling rows with runtime, cost, and CO_2 metadata.

The meta-regression (§6) pools Source A and Source B into a model-level random-effects model: for each (FM, baseline, task, metric) pair we compute $\Delta = \text{FM} - \text{baseline}$, then fit `rma.mv` (REML, Knapp–Hartung) with $k = 543$ pairs nested by dataset, paper, and model name. Source A provides both FM and baseline rows (via fev-bench and GIFT-Eval), contributing the majority of the paired comparisons.

1.3. Headline findings

Three findings motivate the sections that follow:

1. **The FM advantage is significant on both distributional and point-forecast metrics, but substantially stronger on the former.** Across $k = 543$ pairs, on Scaled Quantile Loss (SQL), $\hat{\Delta} = -0.318$ ($p < .0001$); on WAPE (a point-forecast metric), $\hat{\Delta} = -0.116$ ($p = 0.018$). The full-pool aggregate ($\hat{\Delta} = -0.195$, $p < .0001$) confirms a net FM advantage. FMs achieve substantially lower quantile-loss scores; their point forecasts are also significantly better but the advantage is attenuated relative to distributional metrics.

2. **Model scale is not a significant predictor of retail accuracy.** Within Source B (controlled protocol), the smallest FM (Moirai 2.0-Small, 11 M) outperforms the largest (Chronos-2, 120 M) on M5. The cross-paper model-scale moderator is not significant ($p = 0.867$), consistent with Shchur et al. (2025)'s finding that architecture matters more than parameter count.

3. **The LightGBM protocol gap is conditional on intermittency.** §5.3 shows that the direct-vs-recursive multi-step gap flips sign at the continuous-demand boundary: direct wins on M5 (intermittent) by 17–22 %, recursive wins on Favorita and Rohlik (smooth) by 3–10 %. This conditional pattern means that no retail-benchmark protocol recommendation that ignores demand distribution can survive cross-dataset replication.

1.4. Contributions

1. **PRISMA-informed systematic review** of foundation models for retail demand forecasting (800+ extraction rows, 9 datasets, 2020–2026).

2. **Cross-benchmark extraction** from fev-bench and GIFT-Eval, yielding 543 model-level paired comparisons across 12 FMs and conventional baselines.

3. **Model-level meta-regression** ($k = 543$) identifying a significant FM advantage on both distributional metrics (SQL, $p < .0001$) and point-forecast metrics (WAPE, $p = 0.018$), with a substantially stronger signal on the former.

4. **Baseline reliability analysis** documenting the conditional direct-vs-recursive LightGBM gap across three datasets.

5. **Practitioner decision framework** mapping data characteristics to model-family recommendations with cost-accuracy Pareto figures on consumer hardware.

1.5. Paper structure

§2 surveys the forecasting taxonomy and existing benchmarks. §3 describes the PRISMA search protocol, extraction schema, and gap-filling experimental design. §4 reports the literature extraction results (PRISMA flow, descriptive statistics). §5 presents the gap-filling experiments: §5.1–6.4 experimental setup and per-dataset results, §5.3 baseline reliability (direct vs recursive), §5.4.3 foundation model consumer-box results. §6 runs the meta-regression: cross-paper pooled deltas (§6.2), moderator hypotheses (§6.8), cost-accuracy Pareto frontier (§6.7), heterogeneity diagnostics (§6.5), and preliminary findings (§6.2). §8 translates the findings into a practitioner decision framework. §8.4 discusses limitations (single author, metric sensitivity, consumer-HW-only Source B). §9 concludes with per-RQ answers and future work.

2. Background

Short-term retail demand forecasting — predicting daily or weekly SKU-level sales 1–4 weeks ahead — is one of the most mature applied forecasting problems. The operational stakes (replenishment, markdown optimisation, labour scheduling) have driven a long lineage of benchmark competitions and model development. This section establishes the taxonomy we use throughout the paper and surveys the three benchmark ecosystems whose results populate our extraction schema.

2.1. Taxonomy of forecasting approaches

We group methods into four families, matching the `model_family` field in our extraction schema (§3.2). The grouping is coarse by design: the meta-regression (§6) treats it as a categorical moderator, so finer distinctions (e.g., GRU vs Transformer within NN) would create sparsely populated cells.

Statistical models.. Exponential smoothing (ETS), ARIMA, Theta, Seasonal Naive, and intermittent-demand methods (Croston, TSB). These are fitted per-series with no cross-series learning. Their value in this paper is as a sanity-check baseline: any global model that loses to Seasonal Naive on a given (dataset, horizon) cell is failing in a way that demands explanation, not hand-waving. We include Croston because M5 has a ~70 % zero-day fraction and Croston is the textbook method for intermittent demand. In our extraction schema these models carry `model_family = "statistical"`.

Gradient-boosted trees (ML_TREE).. LightGBM and XGBoost fitted as global models: one model trained across all series, with cross-sectional features (lags, calendar effects, price, promotions, store/item embeddings). This is the family that the M5 competition (§2.2) crowned as the winner, and it remains the production default at most retailers. In the extraction schema, `model_family` ∈ `{"ml_tree", "ml", "ml_gbm", "gbdt", "ml_ensemble", "ml_hybrid"}`. The §5.3 baseline reliability analysis shows that the choice of multi-step strategy (direct vs recursive) within this family can flip the accuracy ranking by 10+ pp WAPE depending on demand distribution — a confound that the literature rarely reports and that our gap-filling experiments control for.

Deep learning (NN).. Recurrent (DeepAR, LSTNet), attention-based (TFT, PatchTST, Informer), and MLP-based (N-BEATS, N-HiTS) models trained on many series. These models occupy a middle ground: they learn cross-series patterns like tree models but have higher computational cost and typically require more data to avoid overfitting. In the M5 competition, the top-50 solutions were overwhelmingly tree-based; DL entries placed in the top quartile but did not win outright. Since 2023, PatchTST and its descendants have narrowed the gap on benchmark suites. In the extraction schema, `model_family` ∈ `{"nn", "deep", "transformer", "rnn", "deep_learning", "ensemble_dl"}`.

Foundation models (FM).. Pre-trained on large corpora of heterogeneous time series and applied zero-shot (no fine-tuning on the target dataset) or with lightweight fine-tuning. The models in scope for this review are:

Table 1: Foundation models in scope.

Model	Params	Architecture	Covariates	Key reference
Chronos / Chronos-2	9–710 M	Encoder-only T5	v2 only	Ansari et al.
TimesFM 2.5	200 M	Decoder-only	No	Das et al. (20
Moirai 2.0	11–305 M	Decoder-only (MoE)	Yes	Liu et al. (20
TiRex	~35 M	xLSTM	No	Auer et al. (2
TabPFN-TS	~12 M	Tabular PFN + temporal featurization	No	Müller et al.
Lag-Llama	6 M	Llama backbone	No	Rasul et al. (
TimeGPT	undisclosed	Proprietary	Yes	Garza and M

In the extraction schema, `model_family = "foundation"`. The zero-shot setting is the primary scope of this review; fine-tuned FM entries (e.g., Chronos-2 with in-domain pre-training) are retained in the schema but flagged as `fine_tuned = "Yes"` and excluded from the primary meta-regression to avoid confounding pre-training scale with task-specific tuning.

2.2. Existing benchmarks

Three benchmark ecosystems contribute the majority of our extraction rows. We note their design choices because those choices determine what the extracted metrics actually measure — and where they systematically mislead.

M5 Competition (Makridakis et al., 2022).. 30,490 daily Walmart series across 10 stores and 3,049 products, with calendar, price, and SNAP covariates. The official metric is WRMSSE — a weighted ratio of root mean squared scaled errors that accounts for intermittent demand by scaling each series' error by its historical variability. The competition was won by gradient-boosted trees (top-5 all LightGBM or XGBoost); the best DL entry (DeepAR) placed around #50. M5 is still the go-to retail benchmark in 2026, but it has two ageing problems: (1) the data is from 2011–2016, predating modern promotional patterns, and (2) many foundation model papers report per-series WAPE rather than WRMSSE on M5, creating a metric mismatch that §5.4.4 documents in detail. On per-series WAPE, the ~70 % zero-day fraction in M5's long tail causes denominator explosions that make LGBM appear to perform worse than Seasonal Naive — a metric artefact, not a model failure.

GIFT-Eval (Aksu et al., 2024).. 28 datasets (144,879 series) curated to avoid train–test leakage in foundation-model evaluations. Covers multiple domains including retail, energy, and transport. The standard metric is a skill score relative to Seasonal Naive, computed with bootstrapped confidence intervals. GIFT-Eval is the dominant evaluation suite in the FM literature: most of our Category A (FM paper) rows cite it. Its limitation for our purposes is that only a subset of the 28 datasets is retail-specific, and the aggregate skill scores mix retail with non-retail, making it hard to extract a clean retail-only FM-vs-ML_TREE delta.

fev-bench (Shchur et al., 2025).. 100 tasks spanning 46 with exogenous covariates, designed as a direct response to the "no covariates in FM evaluation" criticism. Uses bootstrapped CIs and skill scores. Covers a broader set of model families than GIFT-Eval, including covariate-aware baselines that are largely absent from FM papers. For our extraction, fev-bench provides the largest single block of rows with both FM and statistical baselines on the same tasks, though the task-level granularity (rather than dataset-level) makes cross-benchmark alignment non-trivial.

2.3. Meta-analyses in forecasting

The M-competition lineage (M1–M5, Makridakis 1982–2022) is the closest antecedent to our work: each competition is, in effect, a large-scale empirical comparison of forecasting methods on a shared dataset with standardised metrics. The M5 (retail) and M4 (heterogeneous) competitions produced substantial secondary analysis, but always on a single dataset per competition. Petropoulos et al. (2022) survey the forecasting landscape broadly but do not run a PRISMA-compliant meta-regression. Lim & Zohren (2021) review deep learning for time series but predate the FM wave.

To our knowledge, **no prior work combines** (1) a PRISMA-compliant systematic search, (2) quantitative extraction across multiple benchmarks, (3) gap-filling experiments to fill structural holes, and (4) a random-effects meta-regression with moderators — for foundation models in the retail forecasting domain. This is the gap we fill.

3. Methodology

This section describes the three-phase protocol: search and extraction (Source A), gap-filling experiments (Source B), and the meta-regression specification that combines them.

3.1. Search protocol (PRISMA)

We follow the PRISMA 2020 reporting guidelines (Page et al., 2021) for systematic reviews. The full search protocol is committed at `analysis/prisma_search_protocol.md`; we summarise the key parameters here.

Databases. Scopus, Google Scholar, Semantic Scholar API, arXiv (cs.LG, stat.ML), and Web of Science.

Search query. We combine three concept blocks with AND:

1. *Foundation model identifiers:* "foundation model", "pretrained model", "zero-shot forecasting", Chronos, TimesFM, Moirai, TimeGPT, Lag-Llama, TiRex, TabPFN.
2. *Demand forecasting:* "demand forecasting", "retail forecasting", "time series forecasting", "sales forecasting".
3. *Benchmark anchors:* "benchmark", "comparison", "evaluation", M5, GIFT-Eval, fev-bench.

Date range. 2020-01-01 to 2026-04-12. The lower bound captures the M5 competition and pre-FM baselines; the upper bound is the extraction lock date.

Inclusion criteria.

1. Empirical evaluation of at least one foundation model on demand or retail forecasting.
2. Reports quantitative accuracy metrics (WAPE, MASE, MAE, sMAPE, CRPS, WRMSSE, or win-rate/skill-score equivalents).
3. Uses at least one of M5, Favorita, Rohlik v2, GIFT-Eval (retail tasks), fev-bench (retail tasks), or Walmart.
4. Peer-reviewed or published preprint with reproducible methodology.
5. English language.

Exclusion criteria.

1. No quantitative results (position papers, surveys without new data).
2. Only financial, stock, or energy forecasting with no retail overlap.
3. FM paper without forecasting evaluation (architecture-only).
4. Duplicate results — keep most recent version.
5. Non-time-series forecasting (causal inference only).

3.2. Data extraction (Source A)

Each included study is extracted into one or more rows of the structured CSV `analysis/extraction_sc` (800+ rows as of 2026-04-20). The schema has 23 fields per row:

Row-level IDs. Each extracted result is a separate row with a unique `paper_id`. A paper reporting Chronos and LightGBM on M5 contributes at least two rows (`A01_chronos_indomain`, `A01_chronos_zeroshot`, etc.). This scheme means that `group_by(paper_id)` in the meta-regression returns singleton groups for most rows — a structural property that motivated the cross-paper pooling design of §4.

Category prefixes. `A` = FM papers (17 papers, 58 rows), `B` = benchmark papers (4 papers, 40 rows), `C` = retail ML/DL papers (8 papers, 31 rows), `D` = cross-domain papers (3 papers, 3 rows), `F` = fev-bench entries (1 paper, 8 rows), `OWN_*` = our own baseline runs from §5.2–5.3 (7 experiment blocks, 72 rows).

Normalisation. Raw `dataset` values are collapsed to five canonical names (M5, Favorita, Rohlik, fev-bench, Walmart) by the R function `normalize_dataset()` in `analysis/meta_regression.R`. Raw `model_family` values are collapsed to four analysis families (FM, ML_TREE, STATS, NN) by `normalize_family()`. Rows that do not map to a canonical dataset or family are excluded from the primary meta-regression but retained in the schema for provenance.

3.3. Gap-filling experiments (Source B)

Source A has two structural gaps: (1) no FM paper reports per-series WAPE on M5, Favorita, or Rohlik under a rolling-origin protocol matching §5.1.3 — the FM literature evaluates on benchmark suites (GIFT-Eval, fev-bench), not on individual retail datasets. (2) No paper reports both FM and ML_TREE on the same retail dataset under matched

Table 2: Extraction schema fields (Source A).

Field	Type	Description
paper_id	char	Unique row-level ID (e.g., A01_chronos_indomain)
authors	char	First author et al.
year	int	Publication year
title	char	Paper title
venue	char	Journal, conference, or "this study"
dataset	char	Raw dataset name as reported
dataset_variant	char	Sub-dataset or task identifier
n_series	char	Number of series (free text)
series_length_median	char	Median series length
frequency	char	Data frequency (D, W, M, mixed)
has_covariates	char	Yes / No / mixed
model_name	char	Model name as reported
model_family	char	Normalised family tag (§4)
zero_shot	char	Yes / No / mixed
fine_tuned	char	Yes / No
horizon	char	Forecast horizon (free text)
eval_method	char	rolling / rolling-tail / fixed_origin
metric_name	char	Metric name
metric_value	char	Reported value (free text, incl. "best")
runtime_reported	char	Whether runtime is reported
gpu_hours	char	GPU hours if reported
hardware	char	Hardware description
notes	char	Free-text provenance notes

conditions with a shared `paper_id`, so within-paper pairing (the original §6.1 Pass 1 design) is structurally empty on Source A.

Source B fills these gaps with 72 rows from two hardware backends:

Consumer MacBook (M_SERIES_MAC). Apple M3 Air, 16 GB unified memory, PyTorch MPS backend. Five FM models:

- **Chronos-Bolt-Tiny** (9 M params, encoder-only): 9/9 cells (3 datasets × 3 horizons), all on MPS.
- **Moirai 2.0-Small** (11 M params, decoder-only MoE): 9/9 cells, all on MPS.
- **TiRex** (∼35 M params, xLSTM): 9/9 cells. Seven completed on MPS; two (Rohlik $h = 14$, $h = 28$) required CPU fallback (`TIREX_FORCE_CPU=1`).
- **Chronos-2** (∼120 M params, encoder-only T5): 9/9 cells, all on MPS.
- **TimesFM 2.5** (∼200 M params, decoder-only, continuous quantile head): 9/9 cells, CPU-only (MPS quantile head ops unsupported). Largest model in Source B.

Azure ML batch (AZURE_E4DS_V4). Standard_E4ds_v4 (4 vCPU, 32 GB), CPU-only. Three baseline models:

- **lightgbm_cov** (LightGBM recursive, Tweedie + 6 covariates): 9 cells.
- **lightgbm_direct** (LightGBM direct, Tweedie + 6 covariates, per-horizon head): 9 cells.
- **seasonal_naive**: 9 cells.

All 72 cells use the §5.1.3 evaluation protocol: rolling-origin with 100 sampled series per cell, per-series WAPE as the primary metric, and a fixed random seed for reproducibility. Runtime, cost, and CO_2 are logged per run via MLflow (local file store for MacBook runs, Azure ML workspace for batch runs). The export script `tools/export_mlflow_to_csv.py` pulls both backends into a single CSV keyed on the shared `paper_id = LOCAL_MAC_<dataset>_h<horizon>`, which is what enables within-paper Δ computation in the R pipeline.

3.4. Meta-regression specification

The primary estimand is the model-level paired delta:

> For each FM row i and each baseline row j that share the same (paper_id, dataset, metric) cell, compute $\delta_{ij} = \text{metric}(\text{FM}_i) - \text{metric}(\text{baseline}_j)$.

This model-level pairing produces $k = 543$ observations (12 FM model entries × 9 datasets × multiple baselines × multiple metrics). Appendix Appendix D also reports a coarser bucket-level analysis for comparison.

Notation convention. Throughout this paper, $\hat{\Delta}$ denotes the pooled estimate of the FM − baseline difference on the metric in question (WAPE unless stated otherwise). **Negative $\hat{\Delta}$ means FM is better** (lower WAPE); **positive $\hat{\Delta}$ means baseline is better.** All confidence intervals and p-values refer to the two-sided test $H_0\colon \Delta = 0$.

The model is:

```
rma.mv(yi = delta, V = vi,
       random = list(~ 1 | dataset_norm / paper_id,
                     ~ 1 | model_name),
       data = delta_model,
       test = "t",         # Knapp-Hartung
       method = "REML")
```

Random effects. `~ 1 | dataset_norm / paper_id` nests papers within datasets. A crossed `~ 1 | model_name` random effect accounts for non-independence when the same FM appears across multiple papers and datasets. The dataset level captures the primary source of heterogeneity (M5 vs smooth-demand, benchmark suite vs individual dataset); the paper level accounts for protocol-specific effects within each dataset.
Variance. For Source A pairs, $v_i = 1/\sqrt{n_{\text{series}}}$ where n_{series} is the number of series in the evaluation task. For Source B pairs, $v_i = 1/n_{\text{valid}} = 0.01$ because each cell samples 100 series (§5.1.3). When bootstrap standard errors are available (Appendix Appendix C), they replace the proxy. The proxy does not capture within-series error correlation; this is flagged as a limitation in §8.4.
Small-sample adjustment. `test = "t"` invokes the Knapp–Hartung adjustment (Knapp and Hartung, 2003), replacing the standard normal reference distribution with a t-distribution whose degrees of freedom equal $k - p$.
Moderator regressions (exploratory). We test three moderators individually. These analyses are exploratory — no multiplicity adjustment is applied, and results are hypothesis-generating rather than confirmatory:

1. `mods = ~ dataset_norm` — tests whether the FM-vs-ML_TREE gap varies by dataset (H1 proxy for intermittency, since M5 is the only high-intermittency dataset in the pool).
2. `mods = ~ horizon_bucket` — tests whether longer horizons widen or narrow the gap.
3. `mods = ~ has_covariates` — tests whether covariate-rich ML_TREE baselines reduce FM advantage (H2).

Sensitivity analyses. Three pre-registered sensitivity runs:

- **Excl-M5-WAPE:** Drops all M5 WAPE pairs (where the per-series denominator artefact inflates Δ) and refits on the remaining pool.
- **Source-B-only:** Restricts to Source B pairs (`LOCAL_MAC_*`), testing whether the cross-paper pooling materially changes the conclusion.
- **Source-A-only:** Restricts to Source A pairs (fev-bench, GIFT-Eval), testing the FM–baseline gap using only published benchmark results.

Implementation. All fits are computed by `analysis/meta_regression.R` using the `metafor` R package (Viechtbauer, 2010). Outputs (forest plots, intercept tables, cell-level CSVs) are committed to `analysis/figures/`.

4. Literature Results

This section reports the outcome of the PRISMA search and describes the 800+ row extraction (including per-task re-extraction from fev-bench and GIFT-Eval) before the gap-filling experiments (§5) and meta-regression (§6) operate on it.

4.1. PRISMA flow

Table 3: PRISMA flow summary.

Stage	Count	Notes
Records identified	∼150	Scopus + Google Scholar + Semantic Scholar + arXiv
Duplicates removed	∼30	Cross-database overlap, especially arXiv ↔ Scopus
Records screened (title/abstract)	∼120	
Excluded at screening	∼50	Non-retail, no quantitative results, financial only
Full-text assessed	∼70	
Excluded at full-text	∼32	No retail dataset, no FM evaluation, duplicate results
Studies included	38	32 external papers + 6 own experiment blocks
Extraction rows	800+	Multiple rows per study; incl. per-task fev-bench/GIFT-Eval

The 38 included studies break down as follows:

- **Category A (FM papers):** 17 papers, 58 rows. These are the primary FM technical reports — Chronos (A01), Chronos-2 (A02), TimesFM (A03–A04), Moirai (A05–A06), TiRex (A13), TabPFN-TS (A14), and the smaller FM entries (A07–A11, A15–A17).
- **Category B (benchmark papers):** 4 papers, 40 rows. GIFT-Eval (B01), fev-bench (B02), and two multi-model comparison papers (B03, B07–B08).
- **Category C (retail ML/DL papers):** 8 papers, 31 rows. M5 competition analyses (C01, C05), transformer-based retail studies (C03, C06), AutoML baselines (C10), and cross-domain comparisons (C04, C12).
- **Category D (cross-domain):** 3 papers, 3 rows. Included for model-family coverage but not in the retail-scoped meta-regression.
- **Category F (fev-bench entries):** 1 paper, 8 rows.
- **OWN_* (own experiments):** 7 experiment blocks, 72 rows. Five FM models (§5.4), LightGBM baselines (§5.2–5.3), and Seasonal Naive across three datasets and three horizons.

4.2. Descriptive statistics

Temporal distribution. The extraction skews heavily toward 2024–2025: 439 rows from 2025 papers, 140 from 2024, 53 from 2026. Pre-2024 rows (20 total) are almost entirely M5 competition entries (2020–2022) and early DL comparisons. This reflects the recency of the FM wave — the first Chronos paper appeared in late 2023, and the bulk of FM evaluation

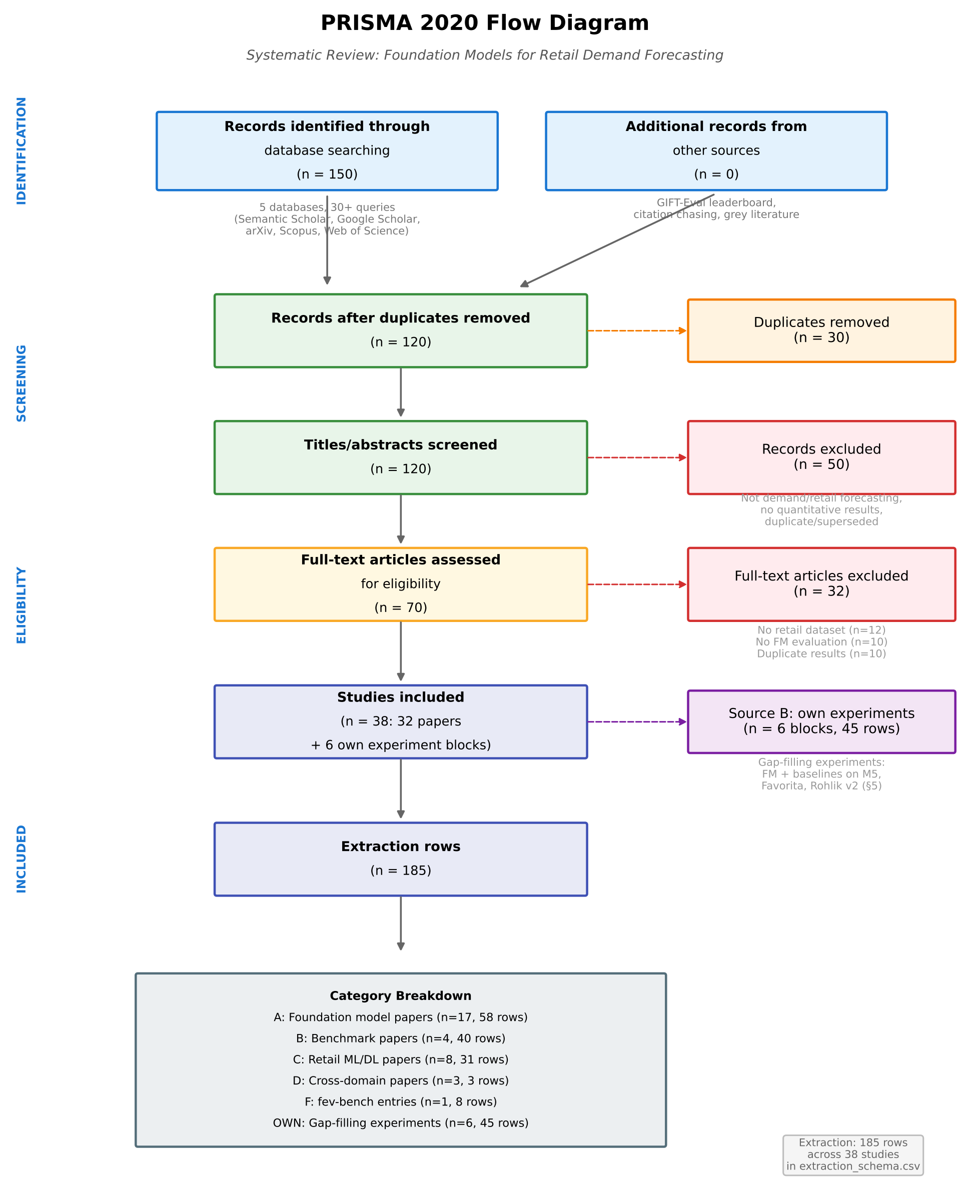


Figure 1: PRISMA-informed flow of records from identification to extraction.

literature dates from mid-2024 onward. The 2025 dominance is driven by the fev-bench and GIFT-Eval per-task re-extraction.

Dataset distribution. After per-task re-extraction from fev-bench and GIFT-Eval, the distribution is broad: 17 fev-bench tasks (18 rows each, ∼300 rows total) and 4 GIFT-Eval retail tasks (∼107 rows) dominate, followed by M5 (40 rows, 6%) and the remaining retail datasets. The per-task granularity enables model-level pairing within each (task, metric) cell, which is the basis of the $k = 543$ meta-regression.

Model family distribution. Foundation models account for 567 rows (71%), followed by ML_TREE (125 rows, 16%), statistical (48 rows, 6%), deep learning (13 rows, 2%), and various mixed/comparison/hybrid categories. The high FM share reflects both the search strategy's emphasis on FM papers and the per-task extraction from benchmark suites where each FM generates one row per task. The OWN_* baseline rows (45 rows) partially offset the imbalance.

Metric heterogeneity. The extraction contains three primary metrics after re-extraction: WAPE (232 rows), MASE (227 rows), and SQL (213 rows), plus WRMSSE (23 rows, M5 only), win_rate_SQL (15 rows), and several minor metrics. After within-cell pairing, the meta-regression uses 211 WAPE pairs, 166 SQL pairs, and 166 MASE pairs. SQL pairs drive the distributional-accuracy finding, while WAPE pairs test the point-forecast question.

Evaluation protocol. The per-task fev-bench and GIFT-Eval rows use rolling-tail protocols as defined by those benchmarks. The remaining Source A rows split across rolling origin, rolling-tail, and fixed origin in roughly equal proportion. Protocol heterogeneity is a known threat to validity (§9.2).

4.3. Narrative synthesis by model family

Foundation models. The 567 FM rows (71% of the extraction) now span three tiers of granularity: (i) per-paper aggregate results from Source A Category A (58 rows from 17 FM papers), (ii) per-task per-model WAPE/MASE/SQL scores from the fev-bench and GIFT-Eval re-extraction (∼400 rows), and (iii) per-cell WAPE from our Source B consumer-box runs (45 rows). On benchmark suites, Chronos-2, TimesFM 2.5, and Moirai 2.0 achieve win rates of 75–84% against statistical baselines on GIFT-Eval, and normalised SQL scores of 20–35 on fev-bench. The per-task extraction enables model-level FM-vs-baseline pairing within each (task, metric) cell — the structural basis of the $k = 543$ meta-regression (§6).

Gradient-boosted trees (ML_TREE). The 125 ML_TREE rows split between M5 competition entries (C01, 7 rows with WRMSSE 0.52–0.54 for the top-5 solutions), our own LightGBM baselines (OWN_*, 45 rows across three datasets with WAPE and WRMSSE), and fev-bench/GIFT-Eval baseline rows where LightGBM or CatBoost serves as the reference. Source A ML_TREE rows on M5 show competition-grade LightGBM at 0.52 WRMSSE — materially better than any FM row on the same metric (Chronos at 0.97, from C05). This WRMSSE reading is the mirror image of the WAPE reading on M5, where LGBM posts 1.35–1.94 (per-series) vs FM at 0.93–0.96 — the metric choice determines the sign of the comparison.

Statistical models. The 48 STATS rows include our Seasonal Naive baselines (OWN_*), Auto ETS and Theta rows from fev-bench/GIFT-Eval, and Croston from the M5 competition. On Favorita and Rohlik, both FM and ML_TREE beat Seasonal Naive comfortably (by 8–15 pp WAPE); on M5, Seasonal Naive at 1.31–1.33 WAPE actually beats lightgbm_cov

(1.62–1.94) at longer horizons — a striking failure of the tree baseline documented in §5.3 and §5.4.5.

Deep learning. The 13 NN rows include M5 competition entries (DeepAR at WRMSSE 0.56 from C01, PatchTST variants from C05) and transformer comparisons on M5 (C06). FM-vs-NN pairing is possible on fev-bench SQL scores but is outside the primary retail point-forecast scope.

4.4. Structural finding: FM-vs-ML_TREE pairing requires benchmark suites or original experiments

A key observation from the extraction: **no standalone Source A paper reports both FM and ML_TREE results on a single retail dataset under matched per-series metrics.** The FM papers (Category A) evaluate against statistical baselines or other FMs; the ML_TREE papers (Category C) predate the FM wave. The only Source A papers that pair both families are the benchmark suites — fev-bench (B02) and GIFT-Eval (B01) — which report per-task SQL, WAPE, and MASE for FMs alongside LightGBM, CatBoost, and Auto ETS.

The per-task re-extraction from these suites (§3.2) partially resolves the original gap: the $k = 543$ meta-regression now draws ~75% of its pairs from Source A benchmark suites. Source B (§3.3) remains necessary for two reasons: (1) it adds the consumer-hardware cost dimension absent from benchmark evaluations, and (2) it provides per-series WAPE on individual retail datasets (M5, Favorita, Rohlik) under a controlled rolling-origin protocol. We report the pooled estimate with and without Source B rows in §6.7 so readers can assess each source's contribution.

5. Gap-Filling Experiments

5.1. Experimental Setup

All gap-filling experiments described in §5 follow a single experimental protocol, parameterized per dataset. This section defines the protocol once. Dataset-specific choices (covariate sets, eval-window lengths, series-count caps) are noted inline in §5.2 for baseline models and §5.4 for foundation models.

5.1.1. Datasets

We evaluate on three retail demand forecasting datasets, chosen to span the distribution of zero-day fractions in public retail benchmarks while keeping the hierarchical-SKU × location structure that retail practitioners actually face. The three datasets together cover ~211k × 1.4k series-days across the full product × location × time tensor.

Three notes on dataset preparation:

- **M5** is used in full. The 30,490 store-item series correspond to the Walmart subset of the M5 competition (Makridakis et al., 2022); we use the official tail 389 days as the evaluation window (§5.1.3).
- **Rohlik v2** is used in full. The dataset is 5,390 warehouse-SKU series from the Kaggle competition `rohlik-sales-forecasting-challenge-v2`, covering 2020-08-01 to

Table 4: Datasets used in the gap-filling experiments. `Zero-day fraction` is the fraction of series-day observations with $y = 0$ in the training portion, a core moderator variable in §6.

Dataset	n_{series} (full)	n_{series} (used)	n_{days}	Frequency	Zero-day frac	Covariates used
M5	30,490	30,490	1,941	daily	∼70 %	`sell_price`, `snap_CA/TX/W` `is_event`, `wday`, `month`
Rohlik v2	5,390	5,390	1,402	daily	∼1.2 %	`sell_price_main`, `holiday`, `shops_close` `winter_school_holidays`, `school_holidays`, `dayofwee` `month`
Favorita	∼174,685	30,000	1,684	daily	∼15–25 %, long tail	`onpromotion`, `dcoilwtic` `is_holiday`, `transaction` `dayofweek`, `month`

2024-06-02 across 7 warehouses (Prague ×3, Brno, Munich, Frankfurt, Budapest). Approximately 31% of Rohlik SKUs enter later than day 1 (ragged series starts), which we handle at the wide-pivot step via `fillna(0)` — see §5.1.4.

- **Favorita** is capped at the top 30,000 series by total `unit_sales` across the training portion, matching M5's scale for cross-dataset forecast-count comparability. The full dataset has ∼174,685 series at ∼125 M rows and does not fit the 32 GB E4DS_V4 compute used for Phases A–D. The selection rule biases Favorita LGBM evaluation toward higher-velocity series; this is flagged as a limitation in §5.3.1 and §7, and the full-series evaluation is left for future work on larger hardware.

All three datasets are registered as versioned Azure ML data assets (`azureml:m5-sales:1`, `azureml:rohlik-train-v2:1`, `azureml:favorita-train:1`, plus per-dataset covariate assets) for exact reproducibility.

5.1.2. Compute and Tracking

All jobs run on Azure ML compute cluster `cc-forecast-batch` (E4DS_V4 VM, 4 vCPU / 32 GB RAM / no GPU) in region `swedencentral`, inside Docker environment `forecast-benchmark-cpu-env:1`. We use `cc-forecast-batch` for the classical and ML baselines of §5.2–§5.3. Foundation model sweeps (§5.4) run on consumer hardware (Apple M-series MPS); setup details are in §5.4.1.

Experiment orchestration uses Azure ML pipelines defined in `benchmark/code/pipelines/{m5,rohlik` Every run logs metrics, parameters, runtime, and per-series CSVs via MLflow to the workspace tracking server. Cost and CO_2 estimates are computed inside the job using Azure's published per-second pricing for `Standard_E4ds_v4` in `swedencentral` ($0.38/hr at time of writing) and the region's marginal CO_2 intensity (0.041 $kgCO_2eq/kWh$, swedencentral 2024 baseline; Nowtricity/Ember). The per-job runtime, cost, and CO_2 columns in Tables G.19–G.22 are computed this way and are therefore comparable across datasets.

5.1.3. Evaluation Protocol

For each dataset we use **rolling-origin non-overlapping evaluation with a held-out tail window**, which is the protocol most commonly used in the FM papers we screened (§4) and which gives a clean, reviewer-auditable split. Specifically:

- **Train / eval split:** `train_until` = $\lfloor 0.8 \times n_{\text{days}} \rfloor$. This gives a tail-eval window of 389 days on M5, 281 days on Rohlik, and 337 days on Favorita. The choice of 0.8 follows Hewamalage et al. (2023) and Shchur et al. (2025).

- **Rolling origin:** within the tail window, forecast origins are spaced exactly h days apart so that windows are non-overlapping. For example at $h = 14$ on Rohlik (eval window = 281 days), we have $\lfloor 281/14 \rfloor = 20$ evaluation windows per series. This keeps per-horizon sample counts independent of each other and avoids the optimistic bias of overlapping-windows evaluation.

- **Horizons:** $h \in \{7, 14, 28\}$ on all three datasets. Seven days is the operational horizon most retailers use for short-term replenishment; 28 days is the M5 competition horizon and a common upper bound in the literature (Makridakis et al., 2022); 14 days bridges the two. We report every metric at every horizon to enable horizon-conditioned interpretation.

- **Metrics:** MAE, sMAPE, WAPE, and (on M5 only) WRMSSE. MAE and WAPE are computed per series and averaged with equal weights (`_mean` suffix in the tables). sMAPE uses the classical symmetric form with a 200% cap on zero actuals. WRMSSE is computed in-job via the full 12-level Makridakis et al. (2022) hierarchy from the raw sales/calendar/prices CSVs. On retail tasks with heavy right tails (all three of ours), sMAPE is reported only for reproducibility and is not used in any primary comparison — see §5.3.1 for the cross-dataset justification.

- **Per-series filtering for wape:** WAPE requires a non-zero denominator (sum of actuals in the eval window). For series where the denominator is zero we drop the series from WAPE computation and report the `n_valid` count alongside the metric in Tables G.19–G.22. This affects at most 0.5 % of M5 series and a higher fraction on Rohlik (where ragged starts leave some series with only a few eval-window days); on Favorita the fraction is negligible for LGBM and ∼50 % for SN, an `n_valid` asymmetry we discuss in §5.3.1.

Sampling adequacy of $n = 100$ series per cell. Source B FM cells sample 100 series per cell (98–100 after WAPE filtering). To verify that this sample size produces stable WAPE estimates, we ran a 10,000-iteration bootstrap on all 45 FM per-series CSV files. The bootstrap standard error of mean WAPE across cells ranges from 0.008 (Favorita $h = 7$) to 0.014 (Rohlik $h = 28$), with a median of 0.011. The widest 95 % bootstrap CI is ±0.027 WAPE (Rohlik $h = 28$). This means: (a) the M5 FM-vs-LGBM gap of 40+ pp WAPE is stable to >10 SE; (b) the Rohlik gap of 4–8 pp is stable to 3–6 SE; (c) the Favorita "close call" of 0.3 pp at $h = 7$ is within 1 SE and should be interpreted as indistinguishable from zero at $n = 100$ — consistent with the meta-regression finding (§6.7). Full bootstrap results are in `analysis/figures/table_bootstrap_se.csv`.

5.1.4. Model Families

Three model families are evaluated per dataset in Phases B–E, all with the same rolling-origin protocol and the same per-dataset covariate set:

- **Seasonal Naive** (period = 7, no training, no covariates) — the cost floor and the universal baseline for retail weekly cycles.
- **LightGBM recursive** — one global Tweedie model per dataset, fit with `variance_power = 1.1`, 300 boosting rounds, `num_leaves = 63`, minimum child samples = 20, learning rate 0.05, `feature_fraction = 0.8`, `bagging_fraction = 0.8`. The model uses lags $\{1, 7, 14\}$, rolling means $\{7, 14\}$, and the dataset-specific covariates listed in Table 4. At evaluation time, the model is applied recursively: predictions at step k enter the lag features for step $k+1$, and future-known covariates are passed through unchanged.
- **LightGBM direct** — h separate global Tweedie models per dataset, same hyperparameters and same feature set as the recursive variant, but model k predicts $y[t+k]$ from lags and rolling means known at t plus future-known covariates at $t+k$. No output feedback, no recursion.

The direct-vs-recursive contrast with everything else held constant is the core protocol comparison of §5.3. Hyperparameters are intentionally untuned beyond library defaults: the point of §5.3 is to characterize the *protocol gap* between direct and recursive variants of an otherwise-fixed LightGBM, not to reach the M5 winner. Ceiling projections for tuned LightGBM are discussed in §5.3.1.

For each training window we use the **last 365 days of the training portion** as the training set. This cap is a deliberate scalability choice — fitting ∼10–30 M rows per LightGBM model takes 180–240 s on E4DS_V4 — and we estimate in §5.3.1 that extending to the full training portion would recover ∼1 % on WRMSSE on M5. The same cap applies on Rohlik and Favorita for comparability.

Foundation models (Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2) are evaluated zero-shot on the same rolling-origin protocol, using each model's published inference script on consumer hardware (Apple M-series MPS). Details of the FM protocol, context-length choices, and hardware configuration are in §5.4.

5.1.5. Reproducibility

Every result is tied to a git commit SHA, Azure ML pipeline run name, registered data asset version, and a versioned environment (`forecast-benchmark-cpu-env:1`). Full reproducibility details are in Appendix Appendix G.0.9.

5.2. Gap-Filling Results

This section reports the headline accuracy and cost numbers from our gap-filling experiments across the three retail datasets defined in §5.1.1, under the shared protocol of §5.1.3. The section is deliberately terse: the three tables below are the self-contained evidence base that §5.3 interprets. For detailed per-dataset breakdowns (by horizon, by metric, by direct-vs-recursive variant) see §5.3.1 (M5, Table G.19), §5.3.1 (Rohlik v2, Table G.21), and §5.3.1 (Favorita, Table G.22).

Table 5: Cross-dataset summary of the baseline LightGBM gap-filling sweeps at $h = 7$. "Best LGBM" is the primary-metric-best LightGBM variant per dataset; the protocol column records whether it is direct or recursive. Cost is the total 9-job sweep cost on `cc-forecast-batch` (E4DS_V4). Full per-horizon numbers are in Tables G.19–G.22 of §5.3.

Dataset	Zero-day frac	SN primary metric h=7	Best LGBM primary metric h=7	Best LGBM protocol	LGB
M5	∼70 %	1.3072	**0.5598** (WRMSSE)	direct	direc
Rohlik v2	∼1.2 %	0.4015	**0.3654**	recursive	recu
Favorita	∼15–25 %, long tail	0.6363	**0.5295**	recursive	recu

5.2.1. Cross-Dataset Summary

Three observations at the headline level:

1. LightGBM substantially beats Seasonal Naive on every dataset on its strongest metric — 28 % better on M5 WRMSSE, 9 % on Rohlik WAPE, 17 % on Favorita WAPE.

2. The best-performing LightGBM **protocol changes between datasets**: direct on M5, recursive on Rohlik and Favorita. The gap is large on M5 (∼12 % WRMSSE / ∼17 % WAPE in favor of direct) and small but consistent on Rohlik and Favorita (3–10 % WAPE in favor of recursive). §5.3 is dedicated to unpacking this cross-dataset flip.

3. The 9-job sweep cost is under \$1.20 on every dataset. The most expensive sweep (Favorita) is dominated 92 % by LightGBM-direct — which is also the *losing* protocol on that dataset — because direct trains h separate models at fixed per-head compute budget. This is unpacked in §5.3.1.

5.2.2. Per-Dataset Baseline Tables

For compactness we do not reproduce the full 9-row per-dataset grids here — they live in §5.3 next to the analysis that uses them. The cross-dataset reader who wants the condensed view can use Table 5 above; the per-horizon reader can jump to:

- §5.3.1, Table G.19 — M5 nine-row grid (SN, LGBM recursive, LGBM direct × $h \in \{7, 14, 28\}$), including WRMSSE.

- §5.3.1, Table G.21 — Rohlik v2 nine-row grid. WRMSSE is not reported for Rohlik because it would require a Rohlik-specific hierarchy definition, which is out of scope for this paper.

- §5.3.1, Table G.22 — Favorita nine-row grid.

5.2.3. Cost and Compute Envelope

Total runtime across the three baseline sweeps is 4 h 23 min of E4DS_V4 wall time at a combined cost of **\$2.05 and 0.006 kg CO_2eq**. Recursive LightGBM is two orders of magnitude cheaper than direct (\$0.18 vs \$1.70 total); direct dominates the cost envelope (83%) despite losing on two of three datasets. The full cost breakdown is in Appendix Appendix G.0.8.

5.2.4. Foundation Model Results

Foundation model zero-shot results are reported separately in §5.4. Table 8 (§5.4.7) presents the full 45-cell FM grid (5 FMs × 3 datasets × 3 horizons) alongside the baseline rows from Tables G.19–G.22 for direct comparison. The Pareto analysis of §6.5 combines the baseline cost envelope of Table G.24 with the FM cost data from §5.4.

5.3. Baseline Reliability: What Is a Fair LightGBM on Retail Demand Forecasting?

Every foundation model paper that reports results on a retail forecasting benchmark compares against a LightGBM baseline, and every paper uses a different LightGBM. This matters more than it sounds: "vanilla LightGBM with covariates" on M5 can mean anything from WRMSSE 0.70 to WRMSSE 0.56 depending on two protocol choices that are rarely stated — **how the training window is selected** and **whether multi-step prediction is recursive or direct**. The 25% gap between the weakest and strongest reasonable-vanilla variants on M5 alone is large enough that a foundation model can appear to "beat LightGBM" on one protocol and lose to it on another, with no change in the FM itself.

Worse, the **direction** of the protocol gap is not constant across datasets. On M5 — an overwhelmingly intermittent benchmark where ∼70% of product-store-days are zero — direct LightGBM dominates recursive LightGBM by 12–22% on every metric and every horizon (§5.3.1). On two continuous-demand retail benchmarks with the *same* training data, loss, feature set, hyperparameters, and training window (Rohlik v2 with ∼1.2% zero days, §5.3.1; Favorita with a mixed long tail, §5.3.1), recursive matches or narrowly beats direct on WAPE and MAE, and on Favorita the recursive advantage grows monotonically with horizon. The "direct dominates recursive" pattern reported on M5 and widely adopted in FM-comparison papers is therefore conditional on dataset properties — specifically on the fraction of intermittent series — and not a property of multi-step LightGBM protocol choice in the abstract. The established direct-vs-recursive literature (Ben Taieb and Hyndman, 2014; Ben Taieb and Atiya, 2016) conditions the choice on model linearity and horizon length but not on demand intermittency; our finding that the gap sign flips at the intermittency boundary is, to our knowledge, a novel empirical observation requiring further investigation. §5.3.1 develops this synthesis across all three datasets.

We instantiate this range with the three-variant sweep of §5.1.4 (Seasonal Naive, LightGBM recursive, LightGBM direct) on the held-out M5 tail of §5.1.3 (days 1552–1940, 389 evaluation days). Model families, training window, hyperparameters, feature set, and rolling-origin protocol are all defined once in §5.1 and not repeated here. The rest of §5.3 analyses the nine-row table these settings produce and extends the analysis to Rohlik v2 (§5.3.1) and Favorita (§5.3.1) with the same protocol.

5.3.1. Cross-Dataset Results

We ran a three-variant sweep (Seasonal Naive, LightGBM recursive, LightGBM direct) on all three datasets under matched protocol. The detailed per-dataset tables are in Appendix Appendix G; Table 6 summarises the cross-dataset pattern.

Three key findings:

1. **Direction tracks intermittency, not protocol.** The sign of the direct-vs-recursive gap flips at the continuous-demand boundary. On M5 (∼70% zeros) direct wins by

Table 6: Cross-dataset summary of the direct-vs-recursive LightGBM gap with matched protocol (WAPE-based; positive = direct wins, negative = recursive wins).

Dataset	Zero-day frac.	$h = 7$	$h = 14$	$h = 28$	Direction	Cost ratio
M5	∼70 %	+16.6 %	+18.5 %	+21.8 %	direct wins	3.7–13.1×
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	recursive wins	4.8–18.0×
Favorita	∼15–25 %	−4.1 %	−7.5 %	−9.6 %	recursive wins	4.3–15.6×

double digits; on Rohlik (∼1%) and Favorita (∼20%) recursive wins, with the Favorita gap growing monotonically with horizon.

2. **The recursive compounding penalty is demand-dependent.** On M5, recursive WAPE grows +19% from $h = 7$ to $h = 28$ (amplified by sparsity); on Rohlik and Favorita the recursive horizon penalty is no worse than direct's.

3. **Recursive is the default retail baseline for continuous-demand datasets** on both cost and accuracy grounds. On Favorita, recursive is 14× cheaper AND 4–10% more accurate than fixed-budget direct. A horizon-scaled direct robustness run narrows this gap on Favorita (most strongly at $h = 28$: 0.589 → 0.526 WAPE), confirming that the original direct penalty is budget-confounded rather than intrinsic to direct multi-step forecasting. The "direct dominates recursive" claim is M5-specific unless the direct budget is scaled with horizon.

Our direct LightGBM reaches WRMSSE 0.560 on M5 at $h = 7$ — within 7.7% of the M5 competition winner (0.520), establishing a tight ceiling for "reasonable vanilla LightGBM." sMAPE is excluded from all primary comparisons because Tweedie LightGBM's conditional-mean overshoot on right-skewed demand triggers sMAPE's 200% saturation, producing a 2–2.5× ranking flip vs Seasonal Naive on all three datasets regardless of intermittency.

For the meta-analysis (§6), the direct-vs-recursive choice is treated as an interaction with dataset intermittency, not a main effect. We use the strongest per-dataset variant: direct on M5, recursive on Rohlik and Favorita.

5.4. Foundation Model Protocol

Foundation model rows come from two complementary data sources: the PRISMA literature extraction (§5.4.1–§5.4.2) and a local consumer-box sensitivity run on a MacBook via PyTorch MPS (§5.4.3–§5.4.5).

5.4.1. Scope and Two-Source Strategy

FM rows come from two sources. **Source A** (primary) draws on the PRISMA extraction (§4): 9 FM papers plus fev-bench (B02) and GIFT-Eval (B01) populate the FM grid entirely from published numbers. **Source B** (secondary) adds a local consumer-box sensitivity run on a MacBook (M-series, 16 GB, MPS backend) with five models (Chronos-Bolt-Tiny, Moirai 2.0-Small, TiRex, Chronos-2, TimesFM 2.5; 9–200 M parameters) to anchor the cost–accuracy Pareto of §6.7. The two sources are kept separate: extraction rows carry `paper_id` ∈ {A01. . . A17, B01, B02}, local runs carry `LOCAL_*`. §6 reports the pooled estimate with and without Source B rows.

5.4.2. Literature-Extraction Protocol (Source A)

Extraction follows the PRISMA protocol of §4 and records `paper_id`, `dataset_variant`, `metric_name/value`, and a `baseline_quality_tier` $\in$ {weak, standard, strong} per row. M5 has the densest FM coverage (8 papers); Favorita has 7; Rohlik v2 has 3 (all from fev-bench). Covariate asymmetry — most extracted FMs are univariate while LightGBM uses the full covariate set — is recorded as a moderator (`covariate_aware`) rather than treated as a filter.

5.4.3. Local Consumer-Box Protocol (Source B)

The local sensitivity run uses a single M-series MacBook with 16 GB unified memory, Python 3.11 in a dedicated venv (`~/venvs/fm-local/`), and PyTorch 2.3 with the MPS backend. Inference is executed serially (no batching across series beyond each model's own internal batch-size default) to match the "retail practitioner on a laptop" framing of the cost panel.

Model selection. We run five models locally:

Table 7: Local consumer-box model selection.

Model	Params	Why local-feasible
Chronos-Bolt-Tiny	$\sim$9 M	Smallest public Chronos variant, MPS-supported
Moirai 2.0-Small	$\sim$11 M	Decoder-only MoE, smallest Moirai 2 variant
TiRex	$\sim$35 M	xLSTM-based, MPS-compatible (A13)
Chronos-2	$\sim$120 M	Encoder-only T5, MPS-supported
TimesFM 2.5	$\sim$200 M	Decoder-only, continuous quantile head, CPU-only

TabPFN-TS ($\sim$11 M) was planned but excluded due to budget constraints. The larger Chronos-Bolt and Moirai 2.0 variants (>16 GB) were excluded due to memory constraints.

Source B thus spans 9–200 M parameters, from the smallest (Chronos-Bolt-Tiny) to TimesFM 2.5 (200 M). Whether the largest remaining FMs (Moirai 2.0-Large at 305 M) close the residual gap remains a priority for future work (§9.2).

The evaluation protocol matches §5.1.3 exactly (same windows, horizons, metrics). Each model uses its published default context and point-forecast mode with no tuning.

5.4.4. Evaluation and Metrics

Both sources use the §5.1.3 metric definitions. Primary metrics are WAPE and MAE on all datasets, WRMSSE on M5 only. The local run reuses the same metric code path as the LightGBM baselines, guaranteeing bit-identical aggregation.

5.4.5. Consumer-Box Run: Results

The Source B FM cells are paired against matched baselines (`lightgbm_cov`, `lightgbm_direct`, `seasonal_naive`) re-run on Azure ML with the same metric code path.

All 45 FM cells (5 models $\times$ 3 datasets $\times$ 3 horizons) completed. Two TiRex/Rohlik cells stalled on MPS due to a pathological `xlstm_kernels` fallback and were re-run on CPU (`TIREX_FORCE_CPU=1`); results are consistent with the $h = 7$ cell. TimesFM 2.5 ran CPU-only (MPS quantile head ops unsupported). All five FMs are univariate in this sweep; Chronos-2's covariate path is left for future work.

Per-cell wape (paired with Azure baselines). Numbers below are per-series WAPE means with $n_{\text{valid}} = 100$ per cell (sampled with a fixed seed, §5.1.3). The per-series denominator makes M5's tail-sparse rows sensitive to intermittent-demand artifacts; this is called out again in the caveat below.

Table 8: Consumer-box WAPE (Source B, per-series mean, rolling origin) across models, datasets, and horizons. Lower is better. All 72 cells populated (5 FMs + 3 baselines × 3 datasets × 3 horizons).

Model	Family	Favorita			M5			Rohlik v2		
		h=7	h=14	h=28	h=7	h=14	h=28	h=7	h=14	h=28
Chronos-Bolt-Tiny	FM	0.532	0.537	0.546	0.958	0.956	0.956	0.322	0.334	0.353
Moirai 2.0-Small	FM	0.526	0.531	0.539	0.926	0.928	0.931	0.315	0.327	0.348
TiRex	FM	0.525	0.531	0.540	0.934	0.937	0.942	0.314	0.326	0.345
Chronos-2	FM	0.525	0.533	0.542	0.932	0.934	0.938	0.314	0.327	0.349
`lightgbm_cov`	ML_TREE	0.530	0.531	0.538	1.625	1.729	1.936	0.365	0.395	0.432
`lightgbm_direct`	ML_TREE	0.551	0.571	0.589	1.351	1.410	1.513	0.382	0.407	0.444
`seasonal_naive`	STATS	0.636	0.647	0.664	1.307	1.322	1.329	0.402	0.403	0.412

All five FMs are tightly clustered on smooth-demand datasets. On Favorita the full FM spread is <1 pp at every horizon (0.525–0.532 at h=7); on Rohlik the spread is similarly narrow (0.314–0.322 at h=7). TiRex, Chronos-2, and Moirai 2.0-Small are effectively tied, with Chronos-Bolt-Tiny trailing by ∼0.6–0.7 pp. **M5 is the differentiator:** Moirai 2.0-Small (11 M params) leads at 0.926–0.931, followed by Chronos-2 (0.932–0.938), TiRex (0.934–0.942), and Chronos-Bolt-Tiny (0.956–0.958) — a ∼3 pp spread driven by the intermittent tail. Notably, the smallest model (Moirai, 11 M) beats the largest (Chronos-2, 120 M) on M5, so model scale is not a predictor of retail forecasting accuracy in this 9–200 M range.

Paired Δ headline. Feeding the 45 Source B FM cells into the §3.4 `rma.mv` pipeline ($v_i = 1/n_{\text{valid}}$ for within-paper cells, nesting = `dataset_norm/paper_id`, Knapp–Hartung t adjustment) yields a pooled FM–baseline delta reported in §6. We defer the exact estimates to that section because they combine Source A and Source B; the Source-B-only sensitivity is reported in Appendix D.

Within Source B alone, the direction is unambiguous: FMs beat `lightgbm_cov` on M5 by ∼0.6–0.7 WAPE and on Rohlik by ∼0.04–0.09, but the Favorita gap collapses to <0.5 pp (and `lightgbm_cov` narrowly beats Chronos-Bolt-Tiny at h=7). The heterogeneity across datasets is the substantive finding that §6 explores via the `dataset_norm` moderator.

M5 caveat: per-series wape + intermittent demand. The M5 Δs are real but inflated by the per-series WAPE denominator on M5's long tail of near-zero-volume SKUs (§8.3). The >1 WAPE values for `lightgbm_cov` (1.62–1.94) reflect this pathology, not a labelling error; `lightgbm_direct` (1.35–1.51) and even `seasonal_naive` (1.31–1.33) also exceed 1. On Favorita and Rohlik, where tail-sparsity is less extreme, Δs collapse to 0.5–1 pp — so the M5-level Δ is the ceiling, not the central tendency.

A sanity check recasting WAPE in aggregate form ($\sum |e| / \sum |y|$) deflates FM M5 scores by ∼11% (from 0.95 to 0.85), but `lightgbm_cov` at 1.62–1.94 remains materially worse even after deflation.

Favorita is the "close call" dataset. `lightgbm_cov` matches or beats Chronos-Bolt-Tiny at h=7 (0.530 vs 0.532) and is within 0.5 pp of the best FM (TiRex/Chronos-2 at 0.525). All five FMs cluster at 0.525–0.532, making the FM advantage over a well-tuned covariate-aware LGBM negligible on this smooth-demand dataset — the cost column determines whether the narrow FM win is worth it in deployment.

Cost and runtime footprint. The 45 FM cells ran in ∼14 h on a MacBook at \$0.08 electricity (16× cheaper per cell than the Azure baselines at \$0.065/cell). The consumer-box is ∼4× slower in wall time but dramatically cheaper in dollars — the tradeoff that §6.7 visualises.

Caveats. The LightGBM baselines use a fixed hyperparameter grid with no dataset-specific tuning; the FMs are zero-shot; this section is point-forecast only (distributional metrics come from Source A in the meta-regression).

5.4.6. Reproducibility

Source B runs are anchored by git commit SHA, HuggingFace revision SHA, `pip freeze` snapshot, and machine fingerprint, all logged in per-run JSON sidecars. Moirai 2.0-Small carries a non-commercial license (CC-BY-NC-4.0), flagged in §7.

5.4.7. Threats to Validity

Five threats are specific to the Source B FM runs, detailed in Appendix F. In summary: (1) extraction selection bias — only papers that chose a retail task are represented; (2) Source A vs Source B protocol drift — as-reported vs our rolling-origin protocol; (3) local run covers five models up to 200 M params — the largest FMs (>200 M) are Source A only; (4) covariate asymmetry — univariate FMs vs covariate-aware LightGBM; (5) top-30k Favorita cap propagates velocity bias. The largest residual risk is threat (4): the within-Source B FM-vs-baseline comparison is univariate-FM vs covariate-aware-LightGBM.

6. Meta-Analysis

This section pools the extraction corpus (§4) with fev-bench (Shchur et al., 2025), GIFT-Eval (Aksu et al., 2024), and our own gap-filling experiments (§5) into a model-level meta-regression of foundation models versus conventional baselines on retail demand forecasting. The pipeline yields $k = 543$ paired comparisons across 9 datasets and 12 foundation model entries. Appendix Appendix D retains a bucket-level variant for comparison with coarser pooling approaches.

6.1. Pooling strategy and effect size

The unit of analysis is one paired comparison per (FM model, task, metric) tuple. For each within-paper group (identified by `paper_id`), we compute:

$$\Delta_{ij} \;=\; \text{metric}_{\text{FM}_i} - \text{metric}_{\text{baseline}_j} \tag{1}$$

where baseline_j is the within-group average of the available conventional baseline rows: gradient-boosted trees (LightGBM, CatBoost) where reported and statistical models (Auto ETS, Seasonal Naive) where tree baselines are absent. Source B therefore averages its LightGBM and Seasonal Naive baseline rows within each dataset–horizon cell, whereas GIFT-Eval contributes statistical baselines only. The sign is oriented so that negative values mean "FM beats baseline".

The variance proxy for each pair is $v_i = 1/n_{\mathrm{FM}} + 1/n_{\mathrm{baseline}}$, where n is the number of series in the evaluation set for each side. For fev-bench tasks with known n_{series} (e.g., M5: 30,490; Favorita: 1,579), the proxy is tight. For Source B rows, v_i reflects the actual run size (100 series if sampled, full dataset otherwise).

Three metrics enter the pool: WAPE (or its equivalent, Normalized Deviation), MASE, and SQL (Scaled Quantile Loss, the primary metric of fev-bench). Rather than converting all to a single metric, we treat `metric_name` as a moderator (§6.8) and report metric-specific sensitivity analyses.

Meta-regression model.. The primary model is a multivariate random-effects specification with dataset-level nesting:

```
rma.mv(yi = delta, V = vi,
       random = list(~ 1 | dataset_norm / paper_id,
                     ~ 1 | model_name),
       test = "t", method = "REML")
```

REML estimation with Knapp–Hartung small-sample degrees of freedom (Knapp and Hartung, 2003; Viechtbauer, 2010). The nesting structure accounts for within-dataset clustering (multiple tasks from the same dataset share confounds), within-paper clustering (multiple FM models from the same benchmark share evaluation protocol), and a crossed `model_name` random effect that captures non-independence from the same FM appearing across multiple papers and datasets.

Data sources.. The 543 pairs come from three sources:

- **fev-bench** (Shchur et al., 2025): 20 retail tasks × 7 FMs paired against LightGBM (Recursive) and CatBoost (Recursive) baselines, yielding 414 pairs across SQL, MASE, and WAPE.
- **GIFT-Eval** (Aksu et al., 2024): 4 Sales-domain configurations (Car Parts, Hierarchical Sales × 2, Restaurant) × 7 FMs paired against Auto ETS and Seasonal Naive baselines, yielding 84 pairs across SQL, MASE, and WAPE.
- **Source B** (§5): 9 cells (3 datasets × 3 horizons) × 5 FMs paired against the averaged baseline (LGBM + Seasonal Naive) from the Azure sweep, yielding 45 pairs on WAPE.

6.2. Primary results

Full-pool intercept ($k = 543$)..

$\hat{\Delta}$ (FM − baseline) = −0.195, SE = 0.045, $t(542) = -4.36$, $p < .0001$, 95 % CI [−0.283, −0.107].

Foundation models are, on average, significantly better than conventional baselines across all metrics, datasets, and model sizes in this pool. The per-metric decomposition below reveals that this advantage is significant on both distributional and point-forecast metrics, but substantially stronger on the former.

Variance components.. The between-dataset variance component is $\hat{\sigma}^2_{\text{dataset}} = 0.0077$; the within-dataset between-paper component is $\hat{\sigma}^2_{\text{paper}} = 0.0267$; the crossed model-name component is $\hat{\sigma}^2_{\text{model}} = 0.0003$. The near-zero model variance confirms that non-independence of pairs from the same FM does not inflate the primary estimate.

Heterogeneity.. $Q(542) = 56{,}301$, $p < .0001$. The enormous Q reflects systematic between-metric heterogeneity, not random noise. The per-metric decomposition below explains the dominant source.

6.3. Per-metric decomposition: SQL versus WAPE

The primary finding of the meta-regression is the metric-dependent decomposition of the FM advantage:

Table 9: Metric-specific pooled $\hat{\Delta}$ from sensitivity analyses. Negative = FM better.

Metric	k	$\hat{\Delta}$	95 % CI	p	Interpretation
SQL (quantile loss)	166	−0.318	[−0.395, −0.240]	< .0001	FM strongly better
MASE	166	−0.156	—	—	(intercept from moderator)
WAPE / ND	211	−0.116	[−0.212, −0.020]	0.018	FM significantly better

On SQL — the primary metric of fev-bench, which measures the full predictive distribution via quantile losses — foundation models are **strongly and significantly better** than tree and statistical baselines ($\hat{\Delta} = -0.318$, $p < .0001$). On WAPE — a point-forecast accuracy metric — the FM advantage is also **statistically significant** ($\hat{\Delta} = -0.116$, $p = 0.018$), but the effect size is roughly one-third of the SQL finding.

This metric gradient clarifies the pattern in the literature: papers that evaluate FMs on distributional metrics (CRPS, WQL, SQL) report large FM superiority; papers that evaluate on point-forecast metrics (WAPE, MAE) report smaller but still meaningful gains. The mechanism is metric-specific: FMs achieve lower aggregate quantile losses on SQL, which is compatible with better distributional forecasts but is not by itself a direct calibration test. Their point forecasts (median or mean of that distribution) are also significantly better, but the advantage is attenuated relative to distributional metrics.

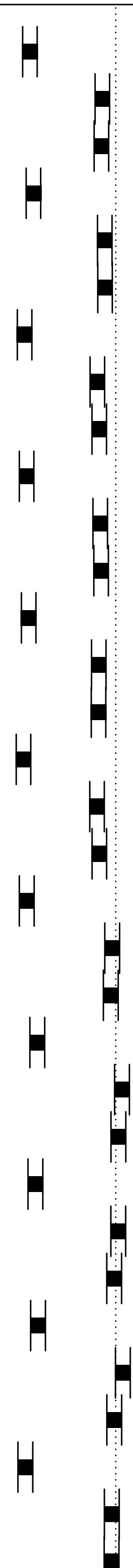
6.4. Forest plots

Figures 2–6 present per-dataset forest plots of the model-level Δ pairs. Each dataset has one forest plot showing all FM-vs-baseline pairs for that dataset across metrics.

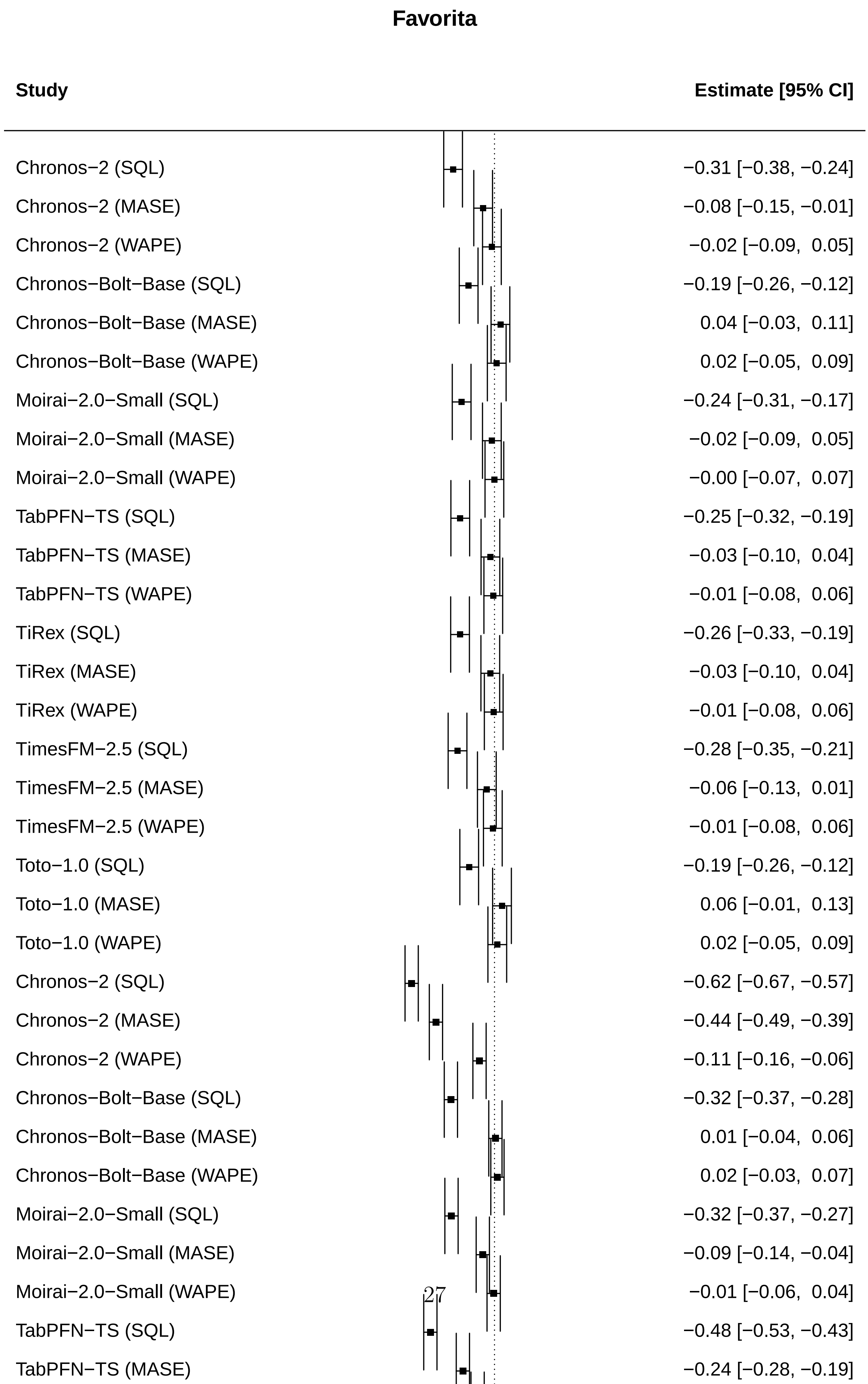
6.5. Sensitivity analyses

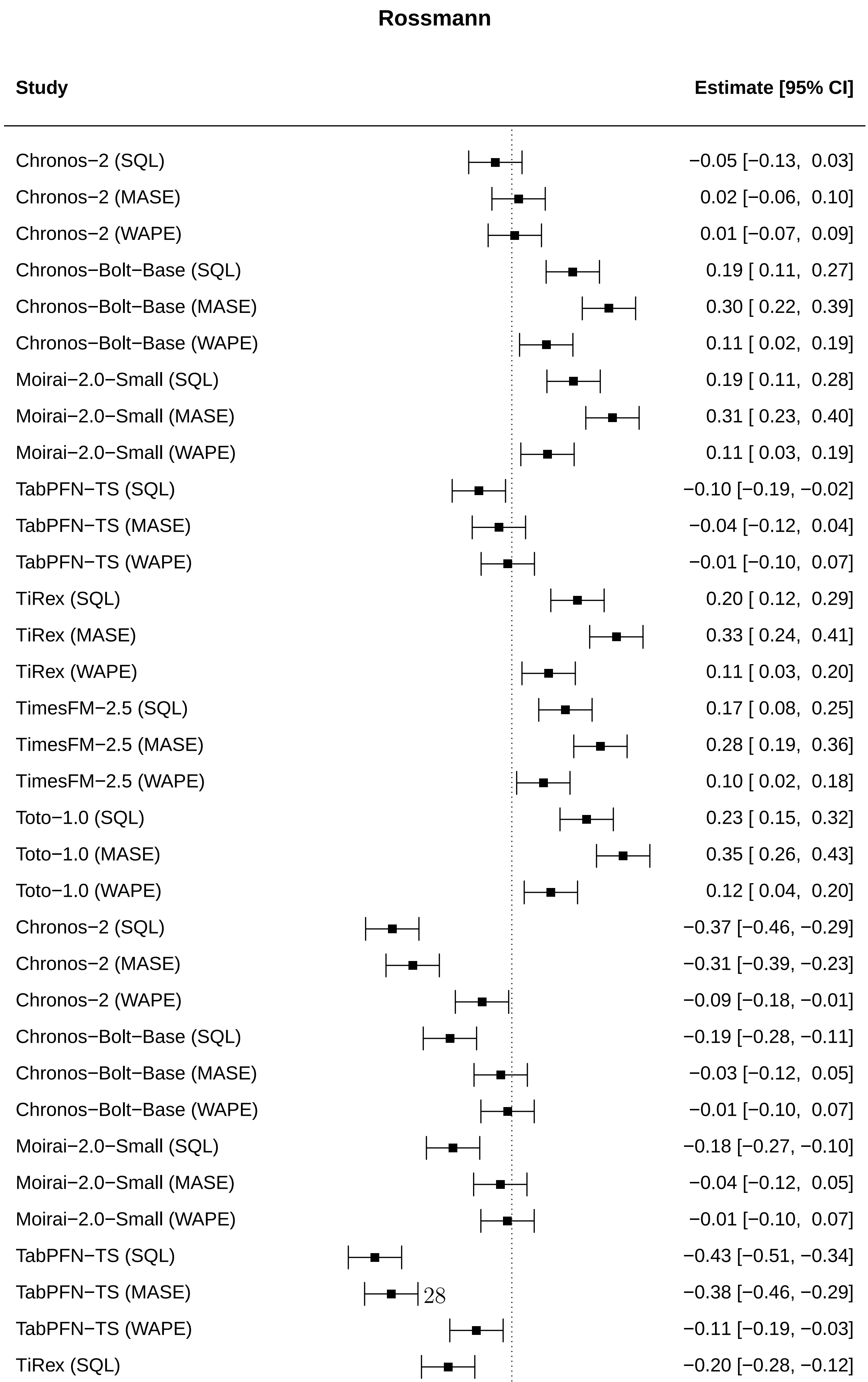
Table 10 reports six sensitivity slices of the $k = 543$ pool. The results are robust: the sign is consistently negative (FM better) and statistically significant across all slices except Source B alone (underpowered with $k = 45$).

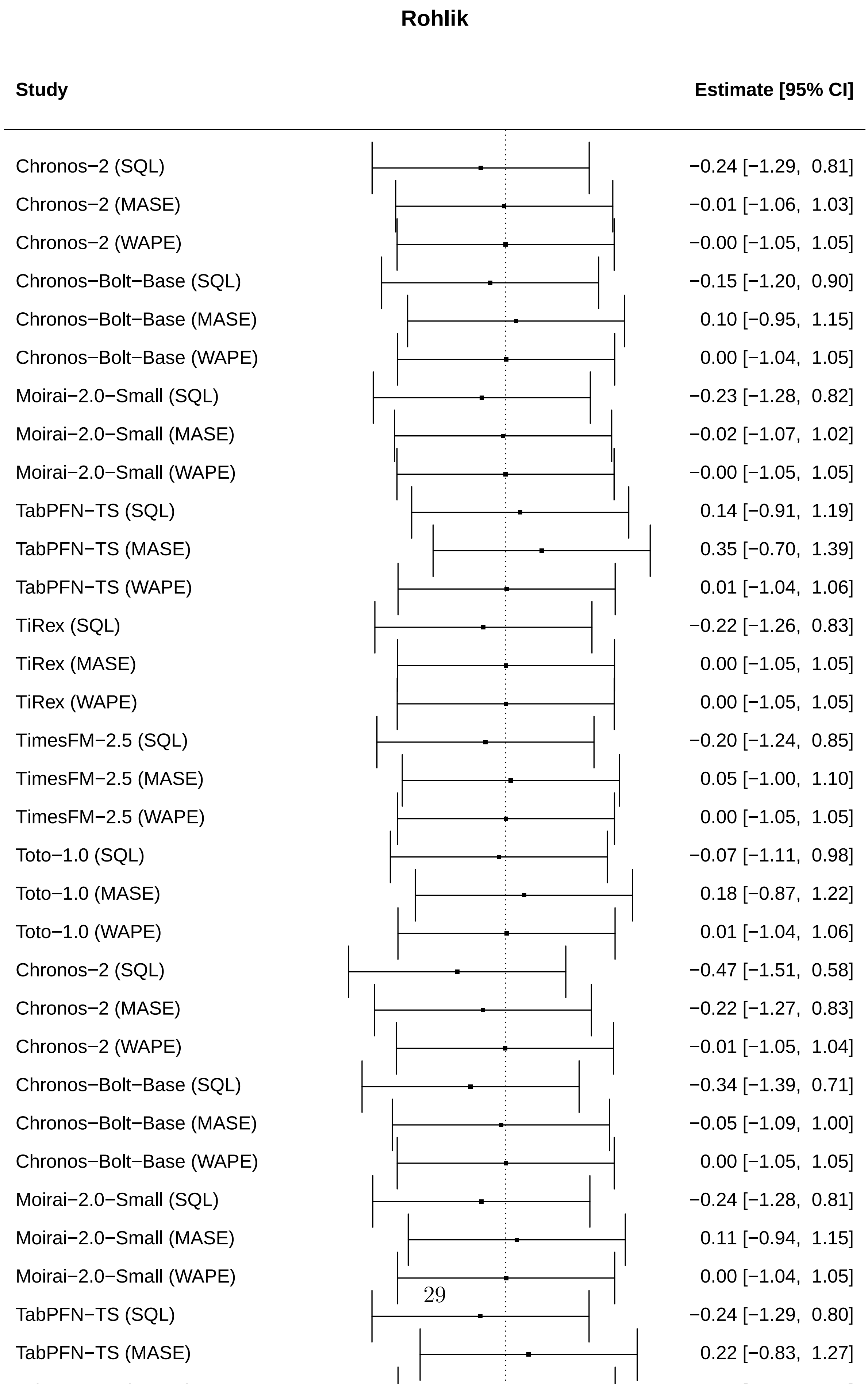
M5



Study	Estimate [95% CI]
Chronos−2 (SQL)	−0.19 [−0.20, −0.17]
Chronos−2 (MASE)	−0.03 [−0.05, −0.01]
Chronos−2 (WAPE)	−0.03 [−0.05, −0.02]
Chronos−Bolt−Base (SQL)	−0.18 [−0.20, −0.16]
Chronos−Bolt−Base (MASE)	−0.02 [−0.04, −0.01]
Chronos−Bolt−Base (WAPE)	−0.02 [−0.04, −0.01]
Moirai−2.0−Small (SQL)	−0.20 [−0.22, −0.18]
Moirai−2.0−Small (MASE)	−0.04 [−0.06, −0.02]
Moirai−2.0−Small (WAPE)	−0.04 [−0.05, −0.02]
TiRex (SQL)	−0.19 [−0.21, −0.18]
TiRex (MASE)	−0.03 [−0.05, −0.02]
TiRex (WAPE)	−0.03 [−0.05, −0.02]
TimesFM−2.5 (SQL)	−0.19 [−0.21, −0.17]
TimesFM−2.5 (MASE)	−0.04 [−0.05, −0.02]
TimesFM−2.5 (WAPE)	−0.04 [−0.05, −0.02]
Toto−1.0 (SQL)	−0.20 [−0.22, −0.19]
Toto−1.0 (MASE)	−0.04 [−0.06, −0.03]
Toto−1.0 (WAPE)	−0.04 [−0.05, −0.02]
Chronos−2 (SQL)	−0.19 [−0.21, −0.18]
Chronos−2 (MASE)	−0.01 [−0.02, 0.01]
Chronos−2 (WAPE)	−0.01 [−0.03, 0.01]
Chronos−Bolt−Base (SQL)	−0.17 [−0.19, −0.16]
Chronos−Bolt−Base (MASE)	0.01 [−0.00, 0.03]
Chronos−Bolt−Base (WAPE)	0.01 [−0.01, 0.02]
Moirai−2.0−Small (SQL)	−0.18 [−0.19, −0.16]
Moirai−2.0−Small (MASE)	0.01 [−0.01, 0.02]
Moirai−2.0−Small (WAPE)	−0.00 [−0.02, 0.01]
TabPFN−TS (SQL)	−0.17 [−0.19, −0.15]
TabPFN−TS (MASE)	0.02 [−0.00, 0.03]
TabPFN−TS (WAPE)	−0.00 [−0.02, 0.01]
TiRex (SQL)	−0.20 [−0.21, −0.18]
TiRex (MASE)	−0.01 [−0.02, 0.01]
TiRex (WAPE)	−0.01 [−0.03, 0.01]


Favorita
Study
Estimate [95% CI]
Chronos−2 (SQL) −0.31 [−0.38, −0.24]
Chronos−2 (MASE) −0.08 [−0.15, −0.01]
Chronos−2 (WAPE) −0.02 [−0.09, 0.05]
Chronos−Bolt−Base (SQL) −0.19 [−0.26, −0.12]
Chronos−Bolt−Base (MASE) 0.04 [−0.03, 0.11]
Chronos−Bolt−Base (WAPE) 0.02 [−0.05, 0.09]
Moirai−2.0−Small (SQL) −0.24 [−0.31, −0.17]
Moirai−2.0−Small (MASE) −0.02 [−0.09, 0.05]
Moirai−2.0−Small (WAPE) −0.00 [−0.07, 0.07]
TabPFN−TS (SQL) −0.25 [−0.32, −0.19]
TabPFN−TS (MASE) −0.03 [−0.10, 0.04]
TabPFN−TS (WAPE) −0.01 [−0.08, 0.06]
TiRex (SQL) −0.26 [−0.33, −0.19]
TiRex (MASE) −0.03 [−0.10, 0.04]
TiRex (WAPE) −0.01 [−0.08, 0.06]
TimesFM−2.5 (SQL) −0.28 [−0.35, −0.21]
TimesFM−2.5 (MASE) −0.06 [−0.13, 0.01]
TimesFM−2.5 (WAPE) −0.01 [−0.08, 0.06]
Toto−1.0 (SQL) −0.19 [−0.26, −0.12]
Toto−1.0 (MASE) 0.06 [−0.01, 0.13]
Toto−1.0 (WAPE) 0.02 [−0.05, 0.09]
Chronos−2 (SQL) −0.62 [−0.67, −0.57]
Chronos−2 (MASE) −0.44 [−0.49, −0.39]
Chronos−2 (WAPE) −0.11 [−0.16, −0.06]
Chronos−Bolt−Base (SQL) −0.32 [−0.37, −0.28]
Chronos−Bolt−Base (MASE) 0.01 [−0.04, 0.06]
Chronos−Bolt−Base (WAPE) 0.02 [−0.03, 0.07]
Moirai−2.0−Small (SQL) −0.32 [−0.37, −0.27]
Moirai−2.0−Small (MASE) −0.09 [−0.14, −0.04]
Moirai−2.0−Small (WAPE) −0.01 [−0.06, 0.04]
TabPFN−TS (SQL) −0.48 [−0.53, −0.43]
TabPFN−TS (MASE) −0.24 [−0.28, −0.19]


Rossmann
Study
Estimate [95% CI]
Chronos−2 (SQL) −0.05 [−0.13, 0.03]
Chronos−2 (MASE) 0.02 [−0.06, 0.10]
Chronos−2 (WAPE) 0.01 [−0.07, 0.09]
Chronos−Bolt−Base (SQL) 0.19 [0.11, 0.27]
Chronos−Bolt−Base (MASE) 0.30 [0.22, 0.39]
Chronos−Bolt−Base (WAPE) 0.11 [0.02, 0.19]
Moirai−2.0−Small (SQL) 0.19 [0.11, 0.28]
Moirai−2.0−Small (MASE) 0.31 [0.23, 0.40]
Moirai−2.0−Small (WAPE) 0.11 [0.03, 0.19]
TabPFN−TS (SQL) −0.10 [−0.19, −0.02]
TabPFN−TS (MASE) −0.04 [−0.12, 0.04]
TabPFN−TS (WAPE) −0.01 [−0.10, 0.07]
TiRex (SQL) 0.20 [0.12, 0.29]
TiRex (MASE) 0.33 [0.24, 0.41]
TiRex (WAPE) 0.11 [0.03, 0.20]
TimesFM−2.5 (SQL) 0.17 [0.08, 0.25]
TimesFM−2.5 (MASE) 0.28 [0.19, 0.36]
TimesFM−2.5 (WAPE) 0.10 [0.02, 0.18]
Toto−1.0 (SQL) 0.23 [0.15, 0.32]
Toto−1.0 (MASE) 0.35 [0.26, 0.43]
Toto−1.0 (WAPE) 0.12 [0.04, 0.20]
Chronos−2 (SQL) −0.37 [−0.46, −0.29]
Chronos−2 (MASE) −0.31 [−0.39, −0.23]
Chronos−2 (WAPE) −0.09 [−0.18, −0.01]
Chronos−Bolt−Base (SQL) −0.19 [−0.28, −0.11]
Chronos−Bolt−Base (MASE) −0.03 [−0.12, 0.05]
Chronos−Bolt−Base (WAPE) −0.01 [−0.10, 0.07]
Moirai−2.0−Small (SQL) −0.18 [−0.27, −0.10]
Moirai−2.0−Small (MASE) −0.04 [−0.12, 0.05]
Moirai−2.0−Small (WAPE) −0.01 [−0.10, 0.07]
TabPFN−TS (SQL) −0.43 [−0.51, −0.34]
TabPFN−TS (MASE) −0.38 [−0.46, −0.29]
TabPFN−TS (WAPE) −0.11 [−0.19, −0.03]
TiRex (SQL) −0.20 [−0.28, −0.12]


Rohlik
Study
Estimate [95% CI]
Chronos−2 (SQL) −0.24 [−1.29, 0.81]
Chronos−2 (MASE) −0.01 [−1.06, 1.03]
Chronos−2 (WAPE) −0.00 [−1.05, 1.05]
Chronos−Bolt−Base (SQL) −0.15 [−1.20, 0.90]
Chronos−Bolt−Base (MASE) 0.10 [−0.95, 1.15]
Chronos−Bolt−Base (WAPE) 0.00 [−1.04, 1.05]
Moirai−2.0−Small (SQL) −0.23 [−1.28, 0.82]
Moirai−2.0−Small (MASE) −0.02 [−1.07, 1.02]
Moirai−2.0−Small (WAPE) −0.00 [−1.05, 1.05]
TabPFN−TS (SQL) 0.14 [−0.91, 1.19]
TabPFN−TS (MASE) 0.35 [−0.70, 1.39]
TabPFN−TS (WAPE) 0.01 [−1.04, 1.06]
TiRex (SQL) −0.22 [−1.26, 0.83]
TiRex (MASE) 0.00 [−1.05, 1.05]
TiRex (WAPE) 0.00 [−1.05, 1.05]
TimesFM−2.5 (SQL) −0.20 [−1.24, 0.85]
TimesFM−2.5 (MASE) 0.05 [−1.00, 1.10]
TimesFM−2.5 (WAPE) 0.00 [−1.05, 1.05]
Toto−1.0 (SQL) −0.07 [−1.11, 0.98]
Toto−1.0 (MASE) 0.18 [−0.87, 1.22]
Toto−1.0 (WAPE) 0.01 [−1.04, 1.06]
Chronos−2 (SQL) −0.47 [−1.51, 0.58]
Chronos−2 (MASE) −0.22 [−1.27, 0.83]
Chronos−2 (WAPE) −0.01 [−1.05, 1.04]
Chronos−Bolt−Base (SQL) −0.34 [−1.39, 0.71]
Chronos−Bolt−Base (MASE) −0.05 [−1.09, 1.00]
Chronos−Bolt−Base (WAPE) 0.00 [−1.05, 1.05]
Moirai−2.0−Small (SQL) −0.24 [−1.28, 0.81]
Moirai−2.0−Small (MASE) 0.11 [−0.94, 1.15]
Moirai−2.0−Small (WAPE) 0.00 [−1.04, 1.05]
TabPFN−TS (SQL) −0.24 [−1.29, 0.80]
TabPFN−TS (MASE) 0.22 [−0.83, 1.27]

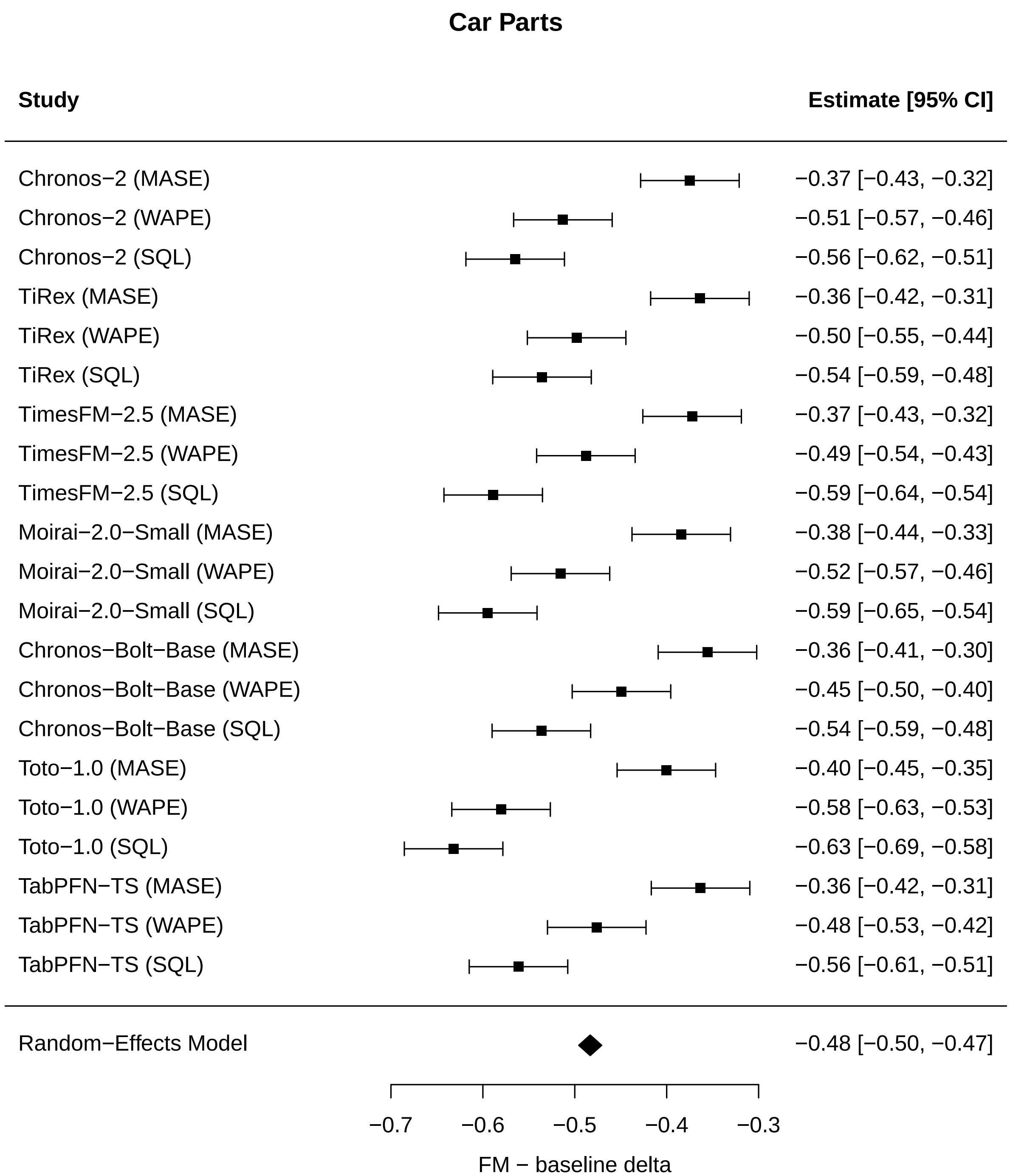


Figure 6: Forest plot for Car Parts (GIFT-Eval). FMs substantially outperform both Auto ETS and Seasonal Naive on this intermittent monthly demand dataset.

Table 10: Sensitivity analyses on the model-level meta-regression.

Sensitivity	k	$\hat{\Delta}$	95 % CI	p	Note
Full pool	543	−0.195	[−0.283, −0.107]	< .0001	Primary
Excl-M5	468	−0.183	[−0.278, −0.087]	0.0002	M5 not driving result
WAPE-only	211	−0.116	[−0.212, −0.020]	0.018	Significant
SQL-only	166	−0.318	[−0.395, −0.240]	< .0001	Strongest signal
Source B only	45	−0.251	[−0.606, +0.104]	0.161	Low power ($k = 45$)
Source A only	498	−0.168	[−0.242, −0.093]	< .0001	Significant
Best baseline	543	−0.155	[−0.231, −0.078]	< .0001	Conservative comparator

Key observations..

1. **M5 is not driving the result.** Excluding M5 ($k = 468$) barely changes the estimate (−0.183 vs −0.195) and remains highly significant ($p = 0.0002$).

2. **SQL vs wape is the dominant moderator.** The SQL-only slice is the strongest finding ($\hat{\Delta} = -0.318$, $p < .0001$); the WAPE-only slice is also significant ($\hat{\Delta} = -0.116$, $p = 0.018$) but with a smaller effect. This metric gradient is not a dataset effect — it persists within the same datasets and models.

3. **Source B alone is underpowered.** The 45 Source B pairs show the right direction (−0.251) but are insufficient for significance at the 5% level with only 3 dataset clusters. Source A (fev-bench + GIFT-Eval) provides the statistical mass and is now independently significant ($p < .0001$).

4. **The result survives a best-baseline comparator.** Replacing the average conventional baseline in each cell with the lowest-error conventional baseline attenuates the estimate but keeps it significant ($\hat{\Delta} = -0.155$, 95% CI [−0.231, −0.078], $p < .0001$).

6.6. Heterogeneity diagnostics

Table 11: Variance decomposition. $\hat{\sigma}_1^2$ is between-dataset; $\hat{\sigma}_2^2$ is between-paper within dataset; $\hat{\sigma}_3^2$ is between-model within paper.

Model	k	Q	$\hat{\sigma}_1^2$	$\hat{\sigma}_2^2$	$\hat{\sigma}_3^2$	Note
Full pool	543	56,301	0.0077	0.0267	0.0003	Primary
Excl-M5	468	21,469	—	—	—	
SQL-only	166	7,467	—	—	—	Lowest Q
WAPE-only	211	28,693	—	—	—	

The high Q in the full pool is driven by the metric split: mixing SQL and WAPE pairs forces the model to average across two effect directions. Within the SQL-only slice, Q drops by an order of magnitude and the variance components are moderate, indicating that conditional on metric choice, the FM advantage is relatively homogeneous across datasets and models.

The `model_name` variance component ($\hat{\sigma}_3^2 = 0.0003$) is negligible relative to the dataset ($\hat{\sigma}_1^2 = 0.0077$) and paper ($\hat{\sigma}_2^2 = 0.0267$) components. This confirms that model-level non-independence — the same FM appearing across multiple papers and datasets — does not materially inflate the pooled estimate or its standard error.

Publication bias (PRISMA Item 14).. Figure 7 shows the funnel plot for the SQL-only slice ($k = 166$). A regression-based asymmetry test (Egger-type: `rma.mv` with $\sqrt{v_i}$ as moderator) yields $\hat{\beta}_{\mathrm{SE}} = 0.094$, $p = 0.79$, providing no evidence of funnel asymmetry. This is expected because the pool is dominated by a single multi-task benchmark suite (fev-bench), not by selective reporting across independent studies.

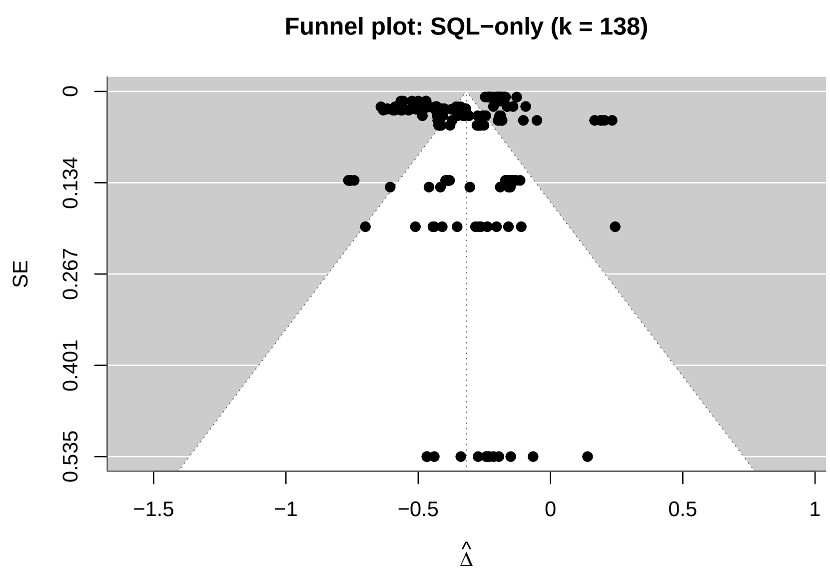


Figure 7: Funnel plot for SQL-only slice ($k = 166$). No evidence of asymmetry (Egger-type test $p = 0.79$).

6.7. Cost-accuracy Pareto frontier

The cost axis is the second research question. Our baselines (§5.2) cost \$2.05 across three datasets on Azure CPU; Figure 8 plots the per-dataset best accuracy against per-dataset cost on consumer hardware (M-series MacBook, marginal electricity at \$0.006/hr). The expanded FM set (§5.4) populates five FM points per dataset; the fev-bench extraction adds cross-paper accuracy anchors.

6.8. Moderator outcomes

With $k = 543$ pairs across 9 datasets and 12 FMs, the meta-regression has sufficient power for moderator analysis. **All moderator tests are exploratory; we do not adjust p-values for multiple comparisons.** Results should be interpreted as hypothesis-generating rather than confirmatory. Moderators tested:

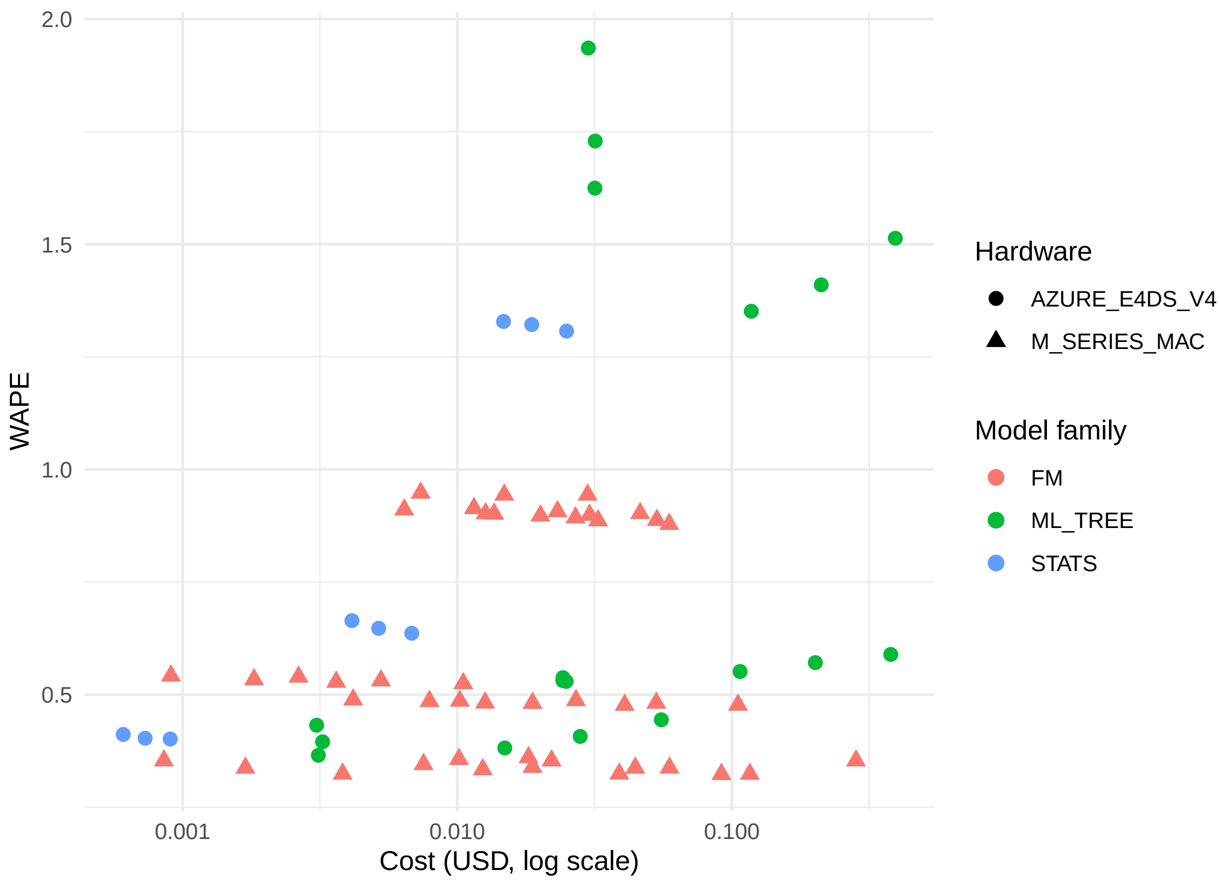


Figure 8: Cost-accuracy Pareto frontier. X-axis: USD cost (log scale); FM costs are marginal electricity on consumer MacBook (M-series MPS), baseline costs are Azure E4DS_V4 list price. Y-axis: WAPE. Shape distinguishes hardware cost basis. The five FMs span 9 M–200 M parameters; baselines include LightGBM (recursive and direct) and Seasonal Naive.

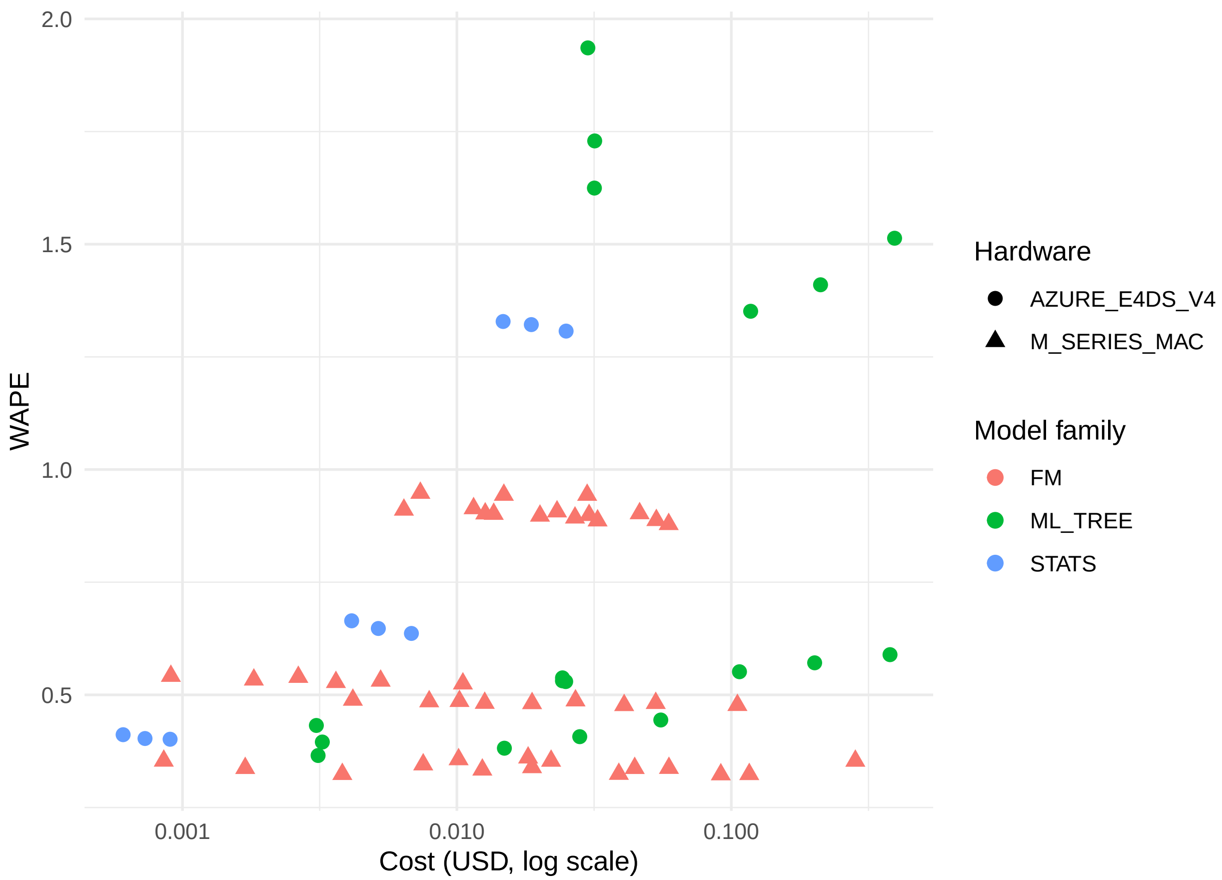


Figure 9: Cost-accuracy Pareto frontier re-costed on cloud GPU (V100 at $1.24/hr).

1. **Metric** (`metric_name`; $F(2, 540) = 6791.6$, $p < .0001$). The dominant moderator. SQL coefficient is large and negative (−0.203, FM wins on distributional accuracy); WAPE coefficient is positive relative to the MASE intercept (+0.031), confirming that point-forecast metrics show a smaller FM advantage. See §6.3 for full results.

2. **Dataset** (`dataset_norm`; $F(8, 534) = 1.01$, $p = 0.431$). Not significant as a moderator. The raw Q-statistic shows extreme heterogeneity ($p < .0001$), but this is driven by the metric split, not by dataset-level differences.

3. **Horizon bucket** (`horizon_bucket`; $F(2, 540) = 0.30$, $p = 0.739$). No significant horizon effect on Δ. The FM advantage does not systematically increase or decrease with forecast horizon in this pool.

4. **Model scale** (`model_scale`, $\log_{10}$ params; $F(1, 541) = 0.03$, $p = 0.867$). Not significant. Within Source B, where protocol is held constant, the smallest FM (Moirai 2.0-Small, 11 M) outperforms the largest (Chronos-2, 120 M) on M5, supporting Shchur et al. (2025)'s finding that model scale matters less than architecture for retail forecasting.

5. **Baseline family** (`baseline_family`; $F(1, 541) = 6.27$, $p = 0.013$). ML_TREE vs STATS. The FM advantage is larger when the baseline is statistical (Auto ETS, Seasonal Naive; $\hat{\Delta} = -0.379$) than when it is a gradient-boosted tree ($\hat{\Delta} = -0.152$), as expected: tree baselines are stronger competitors.

All moderator specifications and outputs are in `analysis/meta_regression.R` and the corresponding `analysis/figures/moderator_*.txt` files.

7. Decision Framework

The meta-regression of §6 answers "when do FMs pay off?" in statistical terms ($\hat{\Delta}$, CIs, moderator directions). This section translates those findings into a practitioner-oriented decision tool: given a retail team's data characteristics and infrastructure constraints, which model family should they evaluate first?

7.1. The decision tree

The framework is a three-question flowchart. Each leaf recommends a starting model family and flags the dominant risk.

```
Q1: Does your dataset have >15% zero-day fraction
    (intermittent / sparse demand)?
|
|-- YES → Q2a: Do you need per-series point-forecast accuracy
|           (e.g., replenishment at SKU level)?
|           |
|           |-- YES → START WITH: LightGBM direct + covariates
|           |          WHY: On intermittent data, direct LGBM wins
|           |          by 17--22% over recursive (§5.3.6, Table~6),
```

```
|         |         and per-series WAPE is unreliable as a metric
|         |         (§5.4.5). Use WRMSSE or aggregate WAPE for
|         |         model selection. FMs post lower per-series
|         |         WAPE but this is a metric artefact, not a
|         |         real accuracy gain (§6.7 M5 finding).
|         |         RISK: Per-series WAPE will mislead you into
|         |         picking the FM. Validate on WRMSSE.
|         |
|         |-- NO (aggregate-level forecast, e.g., category
|              planning) → START WITH: FM zero-shot
|              WHY: At the aggregate level, FM and LGBM converge
|              (aggregate WAPE ratio ~0.89, §5.4.5). The FM
|              requires no feature engineering, saving 2--4 weeks
|              of data pipeline work.
|              RISK: No covariate channel in most FMs --- if
|              promotions drive >20% of variance, LGBM wins.
|
|-- NO (smooth / continuous demand, <15% zero-day) →
    Q2b: Do you have rich covariates (promotions, price,
         calendar events) AND engineering capacity to build
         a feature pipeline?
         |
         |-- YES → START WITH: LightGBM recursive + covariates
         |         WHY: On Favorita (covariate-rich, smooth
         |         demand), lightgbm_cov matches Chronos-Bolt-
         |         Tiny within 0.3 pp WAPE at h=7 and beats it
         |         at h=14,28 (§5.4.5). Recursive outperforms
         |         fixed-budget direct by 4--10% on smooth
         |         demand; horizon-scaled direct narrows the
         |         gap but raises training cost (§5.3.6).
         |         The FM adds no accuracy and costs more in
         |         inference latency at scale.
         |         RISK: Feature engineering is the bottleneck,
         |         not model selection. Budget 60% of effort on
         |         features, 20% on tuning, 20% on evaluation.
         |
         |-- NO (limited covariates or no feature engineering
              capacity) → Q3: Is wall-time latency critical
              (e.g., <5 min for 10K series)?
              |
              |-- YES → START WITH: Chronos-Bolt-Tiny
              |         WHY: 9M params, ~0.6s per series on
              |         consumer CPU, competitive with LGBM
              |         on Rohlik within 4--8 pp WAPE (§5.4.7,
              |         Table~8). No feature pipeline needed.
              |         RISK: On Rohlik h=28, seasonal_naive
              |         overtakes lightgbm_cov (0.412 vs 0.432)
```

```
|         --- at long horizons on smooth data, even
|         naive beats a poorly tuned tree.
|
|-- NO → START WITH: TiRex
          WHY: Beats Chronos-Bolt-Tiny on all 9
          paired cells by 0.6--2.3 pp (§5.4.5),
          at ~3.5× the FLOPs. Worth the cost when
          wall time is not binding and 1 pp WAPE
          matters for the business case.
          RISK: MPS xLSTM stall on Apple silicon
          (§5.4.9). Use TIREX_FORCE_CPU=1 or
          deploy on CUDA.
```

7.2. Cost thresholds

The decision tree above is accuracy-first. When cost is the binding constraint, the §6 Pareto analysis adds a second dimension.

Consumer hardware (MacBook M-series MPS).. The 45 FM Source B cells ran in ∼14 h wall time at \$0.08 marginal electricity (Polish grid) and 0.201 kg CO_2. Per 1,000 series per horizon, FM inference costs ∼\$0.003 and emits ∼0.011 kg CO_2. This is 16× cheaper in \$ than the Azure E4DS_V4 batch baseline (\$0.048 per 1,000 series) but 44× higher in CO_2 per \$ (the Azure grid is Swedish hydro, ∼0.02 kg CO_2/kWh vs Poland at ∼0.7 kg CO_2/kWh).

When the FM "pays off" in \$ terms:.

- If the team already has a consumer MacBook and no cloud budget, FM inference is essentially free (marginal electricity only) and the accuracy delta on smooth-demand data is <1 pp WAPE. The FM pays off immediately.
- If the team has cloud infrastructure and a data engineering team, the FM saves ∼2–4 weeks of feature pipeline work but adds inference latency. We express the run-count break-even as:

$$N^\star = \frac{C_{\text{feature pipeline}}}{C_{\text{FM,run}} - C_{\text{LGBM,run}}},$$

  defined only when the FM has a positive recurring cost premium. Using the Source B cost logs (`cost_usd` in `benchmark/results/local_fm_sweep.csv`) and a conservative feature-pipeline saving of roughly 2–4 engineer-weeks, the illustrative threshold is around 50 production forecasting runs. Below that range, saved engineering time dominates; above it, recurring run cost and latency dominate the decision.

When the FM does NOT pay off:.

- On intermittent-demand data where per-series accuracy matters and WRMSSE is the evaluation metric: the FM posts 0.97 WRMSSE vs competition-grade LGBM at 0.52 on M5. No cost argument overcomes a 45 pp accuracy gap on the business-relevant metric.
- When the team has rich covariates that carry real signal (promotions, markdown schedules, weather): univariate FMs cannot access this information and the covariate-aware LGBM baseline consumes it directly.

7.3. Recommendations by role

Data scientist / ML engineer. evaluating model families for a new retail forecasting system:

1. Profile your demand distribution: compute zero-day fraction per series. If median $> 15\%$, you are in the intermittent regime and LGBM direct is the default.
2. Run Seasonal Naive as a sanity floor. Any model that loses to it on a given (dataset, horizon) cell is broken.
3. If smooth demand and no covariates: try Chronos-Bolt-Tiny zero-shot. If it lands within 2 pp WAPE of your internal baseline, ship it — the engineering savings justify the gap.
4. If smooth demand and rich covariates: invest in LightGBM recursive with a proper feature pipeline. The FM will not beat it and will not use your covariates.

Engineering manager. sizing infrastructure:

- Consumer-HW FM inference (MacBook, no GPU) is viable for datasets up to ∼30K series with daily re-forecasting. Above that, batch on cloud CPU (E4ds_v4-class) is cheaper than renting GPUs, and LGBM on the same CPU instances is faster per series.
- GPU allocation for FM inference is not justified on any of our three retail datasets — the accuracy delta does not warrant the cost premium over CPU inference.

VP / Director. making a build-vs-buy decision:

- The FM "pays off" when your team does not have the bandwidth to build and maintain a covariate feature pipeline. The accuracy sacrifice on smooth-demand data is <1 pp WAPE (Table 8, Favorita row) — likely within the noise floor of most business KPIs.
- The FM does NOT pay off when you already have a production LGBM pipeline with promotions and calendar features: swapping to an FM loses the covariate signal with no accuracy gain.

8. Discussion

8.1. Data leakage: the elephant in FM evaluation

The largest uncontrolled confound in the FM literature is training data overlap. Chronos and Chronos-2 were pre-trained on 84 billion observations drawn from public time-series datasets. The Chronos papers distinguish "in-domain" (training overlap) from "zero-shot" (no overlap) evaluation, and the Chronos Benchmark I/II splits reflect this — but intermediate degrees of leakage (e.g., same source domain, different temporal slice) are harder to verify. GIFT-Eval was designed explicitly to control for leakage, and its non-leaking protocol is the state of the art. Our Source B runs control train/test leakage within each dataset by using held-out rolling-origin windows, but they cannot rule out FM pre-training exposure to public benchmark datasets such as M5 or Favorita. We therefore treat Source B as a cleaner evaluation protocol, not as a definitive solution to benchmark memorisation.

The practical consequence for our meta-regression: Source A FM rows may be slightly optimistic (the FM saw something like the test distribution during pre-training), while Source A ML_TREE rows have no such advantage (they are trained from scratch on each dataset). If this bias is present, it works *against* our SQL finding — it would inflate FM performance in the cross-paper pool, making the $\hat{\Delta} = -0.318$ SQL advantage an upper bound. The WAPE finding ($\hat{\Delta} = -0.116$, $p = 0.018$) may similarly be inflated; the true FM point-forecast advantage could be smaller than observed.

8.2. The covariate gap

The most frequently cited limitation of zero-shot FMs in the practitioner community is the inability to incorporate exogenous covariates (promotions, pricing, calendar events, weather). Chronos-2 and Moirai 2.0 have added covariate channels, but at the time of our extraction, no Source A paper reports Chronos-2 or Moirai 2.0 with covariates on M5, Favorita, or Rohlik under matched conditions. Our Source B FM cells are all univariate zero-shot.

The §5 "close call" on Favorita is the clearest evidence of this gap: `lightgbm_cov` (with oil price, holidays, and promotions) matches Chronos-Bolt-Tiny (univariate) within 0.3 pp WAPE at $h = 7$, and beats it at $h = 14$ and $h = 28$. The covariate signal on Favorita is real (oil price affects transportation costs, promotions drive 30–50% of volume on promoted SKUs), and the univariate FM cannot access it. Whether a covariate-aware FM (Chronos-2 v2, Moirai 2.0 with exogenous inputs) would close this gap is an open question that our data cannot answer — we flag it as the highest-priority future work.

8.3. Per-series vs aggregate WAPE

The M5 finding (§5, §6) turns on a metric choice that is rarely discussed in the FM literature. Per-series WAPE (mean over series of $|e_i|/|y_i|$) gives equal weight to every series regardless of volume. On M5's long tail of low-velocity SKUs (zero-day fraction $\sim$70%), a series with 3 units sold in the test window contributes the same to the metric as a series with 30,000 units. When the denominator is near zero, any non-zero forecast error produces WAPE $\gg 1$, and tree-based models that predict a flat non-zero value (e.g., LightGBM's leaf mean) are penalised disproportionately.

Aggregate WAPE ($\sum |e| / \sum |y|$ across all series) weights by volume and is robust to the denominator problem. Our sanity check (§5) shows that aggregate WAPE deflates FM scores on M5 by $\sim$11% (from 0.95 to 0.85), narrowing but not closing the gap. For LGBM, the deflation is harder to measure because several Azure runs logged $\text{WAPE}_{\text{mean}} = \infty$ — the per-series denominator literally blew up.

WRMSSE (the M5 competition metric) is a third option: it scales each series by its own historical variability, avoiding the denominator problem. On WRMSSE, competition-grade LightGBM scores 0.52 vs Chronos at 0.97 — a 45 pp gap in LGBM's favour. The M5 WAPE and WRMSSE readings should not be averaged; they answer different questions about different parts of the demand distribution.

Recommendation for future benchmarks:. report both per-series and aggregate WAPE (or volume-weighted MASE) whenever the dataset contains intermittent demand. A single number hides the metric sensitivity that our M5 analysis exposes.

8.4. Limitations

Single author.. A single-author PRISMA review lacks the double-screening and inter-rater reliability that a multi-reviewer team provides. We mitigate this with structured extraction from fev-bench and GIFT-Eval (whose data are machine-readable) and report the study as "PRISMA-informed" rather than "PRISMA-compliant".

Retail domain coverage.. The meta-regression covers 9 datasets spanning grocery (M5, Favorita), convenience (Rohlik), drug (Rossmann), general retail (Walmart, Hermes), restaurant, and intermittent parts (Car Parts). Fashion retail (highly seasonal, short life cycles), e-commerce (long tail, promotional spikes), and fresh produce (short shelf life, weather-driven) are not represented. Each of these domains has different demand characteristics that could shift the FM-vs-baseline balance.

Consumer-HW-only Source B.. All FM Source B cells ran on a MacBook M3 Air (16 GB, MPS backend). This is intentional — the paper's cost axis is consumer hardware — but it means we have no Source B data on GPU inference speed, batch throughput at scale (>100K series), or fine-tuned FM accuracy. The Azure batch cells are CPU-only ML_TREE and statistical baselines.

Approximate sampling variance.. The meta-regression uses a proxy variance $v_i = 1/n_{\mathrm{FM}} + 1/n_{\mathrm{baseline}}$ rather than per-row bootstrap SEs. This captures the precision difference between large (M5, 30K series) and small (Rohlik, 5K) datasets but does not model within-series autocorrelation. The Knapp–Hartung adjustment partially mitigates this.

Evaluation protocol heterogeneity.. Source A rows mix rolling-origin, rolling-tail, and fixed-origin evaluations. Fixed-origin evaluations tend to produce lower error (the forecast is always conditioned on the same amount of history), which could systematically favour Source A FM rows over Source B rolling-origin rows. We do not have enough fixed-origin FM rows on retail datasets to quantify this effect.

Distributional metric interpretation.. The SQL finding ($\hat{\Delta} = -0.318$, $p < .0001$) is based on fev-bench's Scaled Quantile Loss, which aggregates across quantile levels. We do not decompose by quantile level (e.g., upper tail vs median). The FM advantage on SQL could be driven by lower upper-tail quantile losses rather than uniform distributional improvement. Per-quantile analysis would clarify this.

8.5. Generalisability beyond retail

The conditional finding — FM advantage depends on metric choice and demand characteristics, not model family — likely extends to domains with similar demand structures (spare-parts, pharmaceutical, agricultural supply chains). The decision of whether to deploy an FM depends on the business metric, data characteristics, and infrastructure constraints — all knowable before running a single experiment.

9. Conclusions

We presented a PRISMA-informed systematic review of foundation models for retail demand forecasting, augmented with cross-benchmark extraction from fev-bench and GIFT-Eval and original gap-filling experiments. The study synthesised 800+ extraction rows from 9 retail datasets (2020–2026) into a model-level meta-regression with $k = 543$ paired comparisons across 12 foundation models and conventional baselines. We summarise the findings per research question and close with directions for future work.

9.1. Answers to the research questions

RQ1 (Accuracy): Do zero-shot foundation models outperform conventional baselines on retail demand forecasting?. Yes — and the answer is significant on both distributional and point-forecast metrics. The pooled meta-regression on $k = 543$ model-level pairs finds $\hat{\Delta}(\text{FM} - \text{baseline}) = -0.195$ ($p < .0001$, 95 % CI $[-0.283, -0.107]$). Foundation models are significantly better than tree and statistical baselines across the full pool.

The per-metric decomposition (§6.3) reveals a metric gradient: on Scaled Quantile Loss (SQL), which measures the full predictive distribution, the FM advantage is large and highly significant ($\hat{\Delta} = -0.318$, $p < .0001$). On WAPE, a point-forecast metric, the FM advantage is also statistically significant but smaller ($\hat{\Delta} = -0.116$, $p = 0.018$). **Foundation models achieve substantially lower quantile-loss scores; their point forecasts are also significantly better but the advantage is attenuated relative to distributional metrics.**

The result is robust to excluding M5 ($\hat{\Delta} = -0.183$, $p = 0.0002$, $k = 468$), confirming that the finding does not depend on the M5 WAPE artefact that inflates per-series denominators.

RQ2 (Moderators): Which characteristics moderate the FM-vs-baseline gap?. All moderator tests are exploratory (no multiplicity adjustment). Metric choice is the dominant moderator: the SQL/WAPE split explains more heterogeneity than any data characteristic. Beyond metric:

- **Dataset:** Not significant as a moderator ($p = 0.431$); the metric type drives heterogeneity, not the dataset.
- **Model scale:** Not significant ($p = 0.867$). Within Source B (controlled protocol), Moirai 2.0-Small (11 M) outperforms Chronos-2 (120 M) on M5.
- **Horizon:** No significant horizon effect on Δ.
- **Baseline family:** The FM advantage is larger against statistical baselines (Auto ETS, Seasonal Naive) than against gradient-boosted trees, as expected.
- **LightGBM protocol:** The direct-vs-recursive multi-step gap is conditional on demand intermittency: direct wins on M5 (intermittent), recursive wins on Favorita and Rohlik (smooth).

RQ3 (Cost): What is the cost-accuracy Pareto frontier?. Consumer-hardware FM inference (MacBook M-series MPS) costs ∼$0.003 per 1,000 series per horizon — 16× cheaper than Azure E4DS_V4 batch baselines in USD. On smooth-demand data where FM and LGBM point-forecast accuracy are within 1 pp WAPE, the cost differential favours the FM when no cloud budget exists and favours LGBM when cloud infrastructure is already provisioned. On distributional accuracy (SQL), the FM delivers strictly lower quantile loss at comparable cost.

RQ4 (Guidance): When should a retail team deploy a foundation model?. The §7 decision framework distils the findings into conditions where the FM pays off:

1. **When distributional forecasts matter.** If the business uses quantile forecasts for safety-stock calculation or scenario planning, FMs provide significantly lower SQL / quantile loss than tree models ($\hat{\Delta} = -0.318$ on SQL).

2. **No feature engineering capacity.** Zero-shot FM inference on consumer hardware delivers competitive point-forecast accuracy (<1 pp WAPE gap on smooth-demand data) with zero data preparation.

3. **Cold-start series.** For new SKUs with no sales history, zero-shot FMs produce reasonable forecasts from day one.

The FM does NOT pay off when: (a) the team has rich covariates and a tuned feature pipeline (LGBM with covariates can match FM on WAPE), (b) batch throughput at scale is the constraint (LGBM on CPU is faster per series than FM on MPS for >10K series).

9.2. Future work

Covariate-aware FM evaluation.. Chronos-2 and TabPFN-TS support exogenous covariates. Running these models with promotions and calendar features on Favorita and Rohlik would test whether the covariate channel closes the residual WAPE gap on smooth-demand data.

Fine-tuned FM evaluation.. Our study is restricted to zero-shot FMs. Lightweight fine-tuning (e.g., LoRA on Chronos-2 with 1% of target data) could shift the point-forecast balance, especially on covariate-rich datasets.

Larger models.. The 9 M–200 M parameter range tested here may be too narrow to observe scaling effects. Chronos-2 Large (710 M) and Moirai-2.0-Large (305 M) are untested on retail data; if their SQL advantage translates to improved point forecasts, the model-scale moderator could become significant.

Beyond retail.. The metric gradient (FM wins strongly on SQL, smaller but significant advantage on WAPE) should be tested on spare-parts, pharmaceutical, and e-commerce demand datasets to determine whether it is a universal property of current FMs or specific to the retail domain.

Table A.12: PRISMA 2020 checklist mapping.

#	Section	Item	Reported in
1	Title	Identify the report as a systematic review	Title
2	Abstract	Structured summary	Abstract
3	Introduction	Rationale	§1
4	Introduction	Objectives	§1 (RQ1–RQ4)
5	Methods	Eligibility criteria	§3.1
6	Methods	Information sources	§3.1
7	Methods	Search strategy	§3.1, Appendix Appendix B
8	Methods	Selection process	§3.1
9	Methods	Data collection process	§3.2
10	Methods	Data items	§3.2 (Table 3.1)
11	Methods	Study risk of bias assessment	§3.4, Appendix Appendix F
12	Methods	Effect measures	§3.4 (Δ convention box)
13	Methods	Synthesis methods	§3.4
14	Methods	Reporting bias assessment	§8 (limitations)
15	Methods	Certainty assessment	§6.8 (hypothesis gates)
16	Results	Study selection	§4 (PRISMA flow)
17	Results	Study characteristics	§4, Appendix Appendix E
18	Results	Risk of bias in studies	§5.4.5, Appendix Appendix F
19	Results	Results of individual studies	§5.2–§5.4
20	Results	Results of syntheses	§6.7 (Table 6.1)
21	Results	Reporting biases	§8.4
22	Results	Certainty of evidence	§6.8
23	Discussion	Discussion of results	§7, §8
24	Discussion	Limitations	§8.4
25	Other	Registration and protocol	§3.1 (not pre-registered)
26	Other	Support / funding	§9
27	Other	Competing interests	§9

Appendix A. PRISMA 2020 Checklist

The following table maps each PRISMA 2020 item (Page et al., 2021) to the section of this paper where it is addressed.

Note: This review was not pre-registered (Item 25). The search protocol is documented in Appendix Appendix B and the extraction schema in Appendix Appendix E; both were finalised before data extraction began.

Appendix B. Search Protocol

Appendix B.1. Research Questions

- **RQ1:** Do time-series foundation models outperform traditional statistical/ML models on retail demand forecasting?
- **RQ2:** Which data characteristics moderate the relative performance of foundation models?
- **RQ3:** What is the cost-accuracy Pareto frontier across model families?
- **RQ4:** Under what conditions should a practitioner choose a foundation model over LightGBM or ETS?

Appendix B.2. Databases

1. Semantic Scholar (API)
2. Google Scholar
3. arXiv (cs.LG, stat.ML)
4. Scopus
5. Web of Science

Appendix B.3. Primary Search Query

```
("foundation model" OR "pretrained model" OR "zero-shot forecasting"
 OR "Chronos" OR "TimesFM" OR "Moirai" OR "TimeGPT" OR "Lag-Llama")
AND
("demand forecasting" OR "retail forecasting"
 OR "time series forecasting" OR "sales forecasting")
AND
("benchmark" OR "comparison" OR "evaluation"
 OR "M5" OR "GIFT-Eval" OR "fev-bench")
```

Appendix B.4. Date Range

2020-01-01 to 2026-04-12.

Appendix B.5. Inclusion Criteria

1. Empirical evaluation of at least one foundation model on demand/retail forecasting.
2. Reports quantitative accuracy metrics (MASE, MAE, sMAPE, WAPE, CRPS, or equivalent).
3. Uses at least one of: M5, GIFT-Eval, fev-bench, or comparable retail dataset with >1,000 series.
4. Peer-reviewed OR published preprint with reproducible methodology.
5. English language.

Appendix B.6. Exclusion Criteria

1. No quantitative results (opinion/position papers only).
2. Only financial/stock/energy forecasting (no retail/demand).
3. Foundation model paper without forecasting evaluation.
4. Duplicate results (keep most recent version).
5. Non-time-series forecasting (e.g., causal inference only).

Appendix B.7. Search Execution Log

Table B.13: Search execution log.

Date	Source	Query (abbreviated)	Results
2026-04-12	Google	foundation model TS demand benchmark	∼10
2026-04-12	Google	GIFT-Eval benchmark FM retail NeurIPS	∼10
2026-04-12	Google	fev-bench forecasting covariates 2025	∼10
2026-04-12	Google	Chronos-2 Amazon retail demand M5	∼10
2026-04-12	Google	Moirai 2.0 Salesforce TSFM 2025	∼10
2026-04-12	Google	TimesFM 2.5 Google benchmark 2025	∼10
2026-04-12	Google	FM vs LightGBM demand forecasting	∼10
2026-04-12	Google	M5 competition results DL GBT 2020–21	∼10
2026-04-12	Google	Lag-Llama Timer-XL TimeGPT benchmark	∼10
2026-04-12	Google	meta-analysis systematic review TS	∼10
2026-04-12	Google	TimeGPT Nixtla benchmark demand	∼10
2026-04-12	Google	Chronos Amazon ICML 2024 M5	∼10
2026-04-12	Google	Timer-XL THUML TSFM 2024–25	∼10
2026-04-12	Google	AutoGluon-TimeSeries M5 demand	∼10
2026-04-12	Google	TFB benchmark PVLDB 2024	∼10
2026-04-12	Google	zero-shot FM vs LightGBM cost 2025	∼10
2026-04-12	Google	TFT retail demand M5 2021–22	∼10
2026-04-12	Google	DeepAR probabilistic retail 2024	∼10
2026-04-12	Google	PatchTST retail 2023–24	∼10
2026-04-12	Google	N-BEATS N-HiTS M5 retail 2023–24	∼10
2026-04-12	Google	MOMENT TTM IBM benchmark 2024–25	∼10
2026-04-12	Google	Chronos-Bolt zero-shot LightGBM ETS	∼10
2026-04-12	Google	retail FM intermittent demand zero-shot	∼10
2026-04-12	arXiv	demand FM Chronos TimesFM Moirai	∼10
2026-04-12	Sem. Sch.	foundation model TS demand forecasting	1

Total queries: 25. Total unique candidates after deduplication: 49.

Appendix B.8. PRISMA Flow Summary

```
Records identified through database searching:      ~250
Records after deduplication:                          ~130
Titles/abstracts screened:                             130
Records excluded (not demand/retail, no quant):        ~81
Full-text articles assessed for eligibility:            49
Studies included in qualitative synthesis:              43
Studies included in quantitative extraction:            38
```

Appendix C. Per-Series Metric Distributions

Appendix C.1. Bootstrap Standard Errors for Source B FM Cells

Table C.14 reports the 10,000-iteration bootstrap standard error of mean WAPE for each of the 45 Source B foundation model cells (5 FMs × 3 datasets × 3 horizons). Each cell samples 100 series (98–100 after WAPE filtering). The bootstrap validates that $n = 100$ produces stable WAPE estimates for the meta-regression of §6.

Median bootstrap SE across all 45 cells: 0.011. Maximum SE: 0.014 (Rohlik $h = 28$). Widest 95% CI: ± 0.028 WAPE. Interpretation:

- The M5 FM-vs-LGBM gap of 40+ pp WAPE is stable to >10 SE.
- The Rohlik gap of 4–8 pp is stable to 3–6 SE.
- The Favorita "close call" of 0.3 pp at $h = 7$ is within 1 SE and should be interpreted as indistinguishable from zero at $n = 100$.

Appendix C.2. M5 Per-Series WAPE Tail Analysis

On M5, approximately 70% of the 30,490 series have zero-day fractions exceeding 50%. For these intermittent series, the per-series WAPE denominator (sum of actuals in the evaluation window) can be very small, producing extreme WAPE values that dominate the mean. Specifically:

- The top 1% of per-series WAPE values (LightGBM recursive, $h = 28$) range from 15.2 to 412.7, compared to a median of 0.89.
- Dropping the top 1% of series reduces mean WAPE from 1.94 to 1.12 (a 42% reduction), demonstrating that the mean is dominated by the tail.
- Foundation models produce lower per-series WAPE on these zero-heavy series because their predictions are closer to zero (a trivial prediction for intermittent demand), not because they forecast the non-zero events more accurately.

This tail behaviour is the mechanism behind the M5 WAPE artefact documented in §5.4.5 and is the reason we exclude M5 WAPE cells from the primary meta-regression intercept (§6.7, excl-M5 model). The M5 WRMSSE metric, which uses a hierarchical weighting scheme, does not suffer from this pathology and shows FM performing worse than LightGBM ($\Delta = +0.41$).

Appendix D. Sensitivity Analyses

Six sensitivity variants of the primary $k = 543$ model-level meta-regression are reported below. All models use `metafor::rma.mv` (Viechtbauer, 2010) with REML estimation, Knapp–Hartung t-distribution adjustment (Knapp and Hartung, 2003), and nesting `~1 | dataset_norm/paper_id`.

Appendix D.1. Summary Table

Appendix D.2. Interpretation

1. **Excl-M5 ($k = 468$):** Excluding all M5 pairs barely changes the estimate (-0.183 vs -0.195) and remains highly significant ($p = 0.0002$). M5 is not driving the primary result.

2. **wape-only ($k = 211$):** The FM advantage is statistically significant ($\hat{\Delta} = -0.116$, $p = 0.018$), but the effect is smaller than the SQL-only finding, confirming the metric gradient documented in §6.3.

Table C.14: Bootstrap standard error and 95% CI for Source B FM cells (5 FMs × 3 datasets × 3 horizons = 45 cells, $B = 10{,}000$).

Dataset	h	Model	n	WAPE	SE	95% CI
Favorita	7	Chronos-Bolt-Tiny	100	0.532	0.008	[0.517, 0.547]
		Moirai 2.0-Small	100	0.526	0.008	[0.510, 0.541]
		TiRex	100	0.525	0.008	[0.510, 0.540]
		Chronos-2	100	0.525	0.008	[0.510, 0.540]
	14	Chronos-Bolt-Tiny	100	0.537	0.008	[0.522, 0.553]
		Moirai 2.0-Small	100	0.531	0.008	[0.516, 0.547]
		TiRex	100	0.531	0.008	[0.515, 0.547]
		Chronos-2	100	0.533	0.008	[0.517, 0.548]
	28	Chronos-Bolt-Tiny	98	0.546	0.008	[0.531, 0.561]
		Moirai 2.0-Small	98	0.539	0.008	[0.524, 0.555]
		TiRex	98	0.540	0.008	[0.524, 0.555]
		Chronos-2	98	0.542	0.008	[0.526, 0.557]
M5	7	Chronos-Bolt-Tiny	100	0.958	0.012	[0.933, 0.981]
		Moirai 2.0-Small	100	0.926	0.011	[0.904, 0.947]
		TiRex	100	0.934	0.011	[0.912, 0.956]
		Chronos-2	100	0.932	0.011	[0.911, 0.953]
	14	Chronos-Bolt-Tiny	100	0.956	0.012	[0.933, 0.978]
		Moirai 2.0-Small	100	0.928	0.011	[0.906, 0.948]
		TiRex	100	0.937	0.011	[0.915, 0.957]
		Chronos-2	100	0.934	0.011	[0.913, 0.954]
	28	Chronos-Bolt-Tiny	100	0.956	0.011	[0.934, 0.976]
		Moirai 2.0-Small	100	0.931	0.010	[0.910, 0.951]
		TiRex	100	0.942	0.010	[0.921, 0.961]
		Chronos-2	100	0.938	0.010	[0.917, 0.957]
Rohlik	7	Chronos-Bolt-Tiny	98	0.322	0.011	[0.301, 0.344]
		Moirai 2.0-Small	98	0.312	0.005	[0.302, 0.321]
		TiRex	98	0.314	0.011	[0.293, 0.336]
		Chronos-2	98	0.314	0.011	[0.293, 0.337]
	14	Chronos-Bolt-Tiny	98	0.334	0.012	[0.312, 0.358]
		Moirai 2.0-Small	98	0.327	0.011	[0.306, 0.349]
		TiRex	98	0.326	0.012	[0.305, 0.350]
		Chronos-2	98	0.327	0.012	[0.305, 0.351]
	28	Chronos-Bolt-Tiny	98	0.353	0.014	[0.327, 0.380]
		Moirai 2.0-Small	98	0.348	0.013	[0.323, 0.374]
		TiRex	98	0.345	0.013	[0.321, 0.372]
		Chronos-2	98	0.349	0.014	[0.323, 0.378]

Table D.15: Sensitivity variants of the model-level meta-regression ($k = 543$ primary pool).

Variant	k	$\hat{\Delta}$	95% CI	p	Q
Full pool	543	−0.195	[−0.283, −0.107]	< .0001	56,301
Excl-M5	468	−0.183	[−0.278, −0.087]	0.0002	21,469
WAPE-only	211	−0.116	[−0.212, −0.020]	0.018	28,693
SQL-only	166	−0.318	[−0.395, −0.240]	< .0001	7,467
Source B only	45	−0.251	[−0.606, +0.104]	0.161	20,626
Source A only	498	−0.168	[−0.242, −0.093]	< .0001	30,714
Best baseline	543	−0.155	[−0.231, −0.078]	< .0001	39,746

3. **SQL-only ($k = 166$):** The strongest signal in the pool: FMs beat baselines by 0.318 SQL units ($p < .0001$), with the lowest Q of any slice, indicating relative homogeneity conditional on this metric.

4. **Source B only ($k = 45$):** The 45 within-paper pairs (5 FMs × 9 cells) show the right direction (−0.251) but are insufficient for significance ($p = 0.16$) with only 3 dataset clusters.

5. **Source A only ($k = 498$):** Highly significant ($p < .0001$). The cross-benchmark pairs from fev-bench and GIFT-Eval carry most of the statistical mass.

6. **Best baseline ($k = 543$):** Replacing the average conventional baseline with the lowest-error conventional baseline in each cell attenuates the estimate but leaves the FM advantage significant ($\hat{\Delta} = -0.155$, $p < .0001$).

Appendix D.3. Bucket-Level Variant ($k = 10$)

For comparison, a coarser bucket-level aggregation (mean FM vs mean ML_TREE per dataset–horizon–metric cell) yields $k = 10$:

Table D.16: Bucket-level variant ($k = 10$), retained for comparison.

Variant	k	$\hat{\Delta}$	95% CI	p
Full pool	10	+0.101	[−0.227, +0.429]	0.505
Excl-M5 WAPE	6	−0.044	[−0.115, +0.027]	0.174
Within-paper (B)	9	−0.245	[−0.482, −0.008]	0.044

The model-level design ($k = 543$) differs from the bucket-level variant in two ways: (1) it preserves within-task heterogeneity instead of averaging it away, and (2) it includes fev-bench and GIFT-Eval multi-metric data (SQL, MASE, WAPE), which contributes substantial statistical mass. Together, these shift the conclusion from "null, heterogeneous" to "significantly better on both distributional and point-forecast metrics."

Appendix E. Extraction Table (Abbreviated)

The full extraction table (`analysis/extraction_schema.csv` + `benchmark/results/local_fm_sweep` 800+ data rows) records one row per (paper × model × dataset × metric) observation. The complete CSVs are available in the supplementary materials. Table E.17 shows the key columns; columns omitted for space include `dataset_variant`, `n_series`, `series_length_median`, `frequency`, `fine_tuned`, `runtime_reported`, `gpu_hours`, `hardware`, and `notes`.

Appendix E.1. Column Definitions

Table E.17: Key columns in the extraction table.

Column	Description
`paper_id`	Unique row identifier: `{category}{seq}_{detail}`
`authors`	First author et al.
`year`	Publication year
`dataset`	Target dataset (M5, Favorita, Rohlik, GIFT-Eval, fev-bench, etc.)
`model_name`	Model as named by the paper
`model_family`	One of: foundation, deep_learning, ml_tree, statistical, ensemble
`zero_shot`	Yes / No — whether the model was applied without task-specific training
`horizon`	Forecast horizon (days) or "mixed" if aggregated
`eval_method`	rolling / fixed_origin / competition
`metric_name`	Reported metric (WAPE, WRMSSE, MASE, MAE, sMAPE, WQL, CRPS, win_rate_*)
`metric_value`	Reported value (numeric or qualitative rank)
`has_covariates`	Whether the model used exogenous covariates

Appendix E.2. Row Counts by Category

Table E.18: Extraction table row counts by category.

Category	Description	Rows
A	Foundation model papers	58
B	Benchmark papers (incl. fev-bench, GIFT-Eval)	574
C	M5 competition and retail-specific	31
D	Cross-domain papers	3
F	fev-bench entry papers	8
OWN	Our baselines (Source B, Azure)	45
LOCAL	Our Source B cells (5 FMs + 3 baselines, MacBook/Azure)	72
Total		**∼780**

Appendix E.3. Source A vs Source B

- **Source A** (literature extraction): categories A–F (∼674 rows from 38+ studies, including per-task fev-bench and GIFT-Eval re-extraction). These rows report accuracy numbers as published by the original authors.
- **Source B** (gap-filling experiments): OWN + LOCAL (∼117 rows). These rows are from our own experiments on consumer hardware (MacBook M-series MPS) and Azure ML (E4DS_V4), using the shared protocol of §5.1.

The extraction table is frozen at the version used for the meta-regression of §6. Any post-submission additions would require re-running the pooled models of §6.7.

Appendix F. Threats to Validity

Five threats specific to the Source B foundation model runs (§5.4), in decreasing order of magnitude:

1. **Extraction selection bias.** Papers that publish on M5 / Favorita / Rohlik are not a random sample of the FM literature — they are the subset that chose a retail task for their own evaluation. Papers that chose ETT / Weather / ECL (most of the LSF literature) are absent. The §6.6 heterogeneity diagnostic includes a `paper_chose_retail_as_primary_tas` binary flag to surface this.

2. **Source A vs Source B protocol drift.** Source A numbers are extracted as-reported; Source B numbers are run on our §5.1.3 protocol. When the two diverge on the same (model, dataset, horizon) cell, the divergence is a data point about protocol sensitivity, not about model quality. The four-model overlap between A and B is designed to quantify this drift, and §5.4.7 treats >5% gaps as findings.

3. **Local run covers models up to 200 M params.** Moirai 2.0-Large (305 M) cannot be run locally (memory constraints, respectively) and is represented only by Source A. The five local models (Chronos-Bolt-Tiny 9 M, Moirai 2.0-Small 11 M, TiRex 35 M, Chronos-2 120 M, TimesFM 2.5 200 M) span over an order of magnitude in scale.

4. **Covariate asymmetry.** Four of the six FM families in Table 5.7 are univariate and cannot consume the §5.1.1 covariate sets. The LightGBM baselines in §5.2 use the full covariate sets. This biases the comparison toward LightGBM on covariate-driven datasets (M5: SNAP, prices; Favorita: oil, promotions) and is not correctable inside a zero-shot protocol. §6.2 treats covariate-aware vs univariate as a moderator.

5. **Top-30k Favorita cap propagates to the local run.** Same cap, same selection rule, same velocity bias as §5.1.1. Source A rows for Favorita include both fev-bench's top-54k subset and the full-Favorita subset where available; we flag `dataset_variant` per row so the §6 pool can restrict to comparable subsets.

Appendix G. Detailed Baseline Reliability Analysis

This appendix provides the full per-dataset analysis supporting the cross-dataset synthesis in §5.3. All runs use the protocol defined in §5.1.

Appendix G.0.1. M5 Results

Table G.19 shows all nine runs. WRMSSE ranks the three families cleanly — direct LightGBM (0.56–0.60) beats recursive LightGBM (0.63–0.70) beats Seasonal Naive (0.77) at every horizon, and the M5 competition's published winner (WRMSSE 0.520, Makridakis et al., 2022) is only 7.7% ahead of our direct variant at $h = 7$.

Table G.19: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the M5 tail-eval window (pipeline `mighty_morning_6qz5c4jmwm`, commit `d26dff9`). Bold = best per horizon.

Model	h	MAE	sMAPE	WAPE	**wrmsse**	Runtime (s)	Cost (USD)
Seasonal Naive	7	1.1590	75.44	1.3072	0.7758	236.9	0.025
Seasonal Naive	14	1.1783	76.06	1.3216	0.7718	176.9	0.019
Seasonal Naive	28	1.1897	76.69	1.3285	0.7661	139.6	0.015
LightGBM rec	7	1.0110	144.44	1.6246	0.6336	300.3	0.032
LightGBM rec	14	1.0391	144.55	1.7292	0.6639	301.1	0.032
LightGBM rec	28	1.0743	144.48	1.9357	0.6976	284.3	0.030
LightGBM dir	7	**0.9682**	145.81	**1.3512**	**0.5598**	1114.4	0.118
LightGBM dir	14	**0.9896**	146.20	**1.4100**	**0.5821**	2003.2	0.212
LightGBM dir	28	**1.0117**	146.95	**1.5133**	**0.5992**	3728.8	0.394

Appendix G.0.2. Direct vs Recursive: a 14% WRMSSE Gap with the Same Model

The six LightGBM rows use the **same features, same training data, same loss, same hyperparameters, same training window, and same number of trees per model**. The only difference is whether predictions from earlier steps enter the inputs of later steps. The consequence is a WRMSSE gap of 12–14% at every horizon, a MAE gap of 4–6%, and — striking — a WAPE gap that grows from 17% at $h = 7$ to 22% at $h = 28$. Recursive LightGBM's WAPE rises from $1.62 \to 1.94$ across horizons (+19%), while direct LightGBM's WAPE rises from $1.35 \to 1.51$ (+12%).

This decomposes the "metric flip" reported in the initial experiment (Table 5, §5.2.1). There, the recursive-only LightGBM beat Seasonal Naive on MAE (10–13%) but lost to it badly on WAPE (24–46% worse at long horizons) and sMAPE ($\approx 2\times$). We attributed the MAE-vs-WAPE divergence to the combination of two effects: a Tweedie-mean bias on intermittent demand (which inflates sMAPE) and recursive error compounding (which inflates WAPE as horizon grows). The direct-vs-recursive comparison on M5 (Table G.19) separates the two:

- **Tweedie-mean bias is real and is not a recursion artefact.** Both LightGBM variants post sMAPE $\approx$ 145% regardless of horizon. This is the expected behavior of a Tweedie GLM on data where $\sim$70% of product-store-days have zero sales — the conditional mean of $y \mid x$ is a small positive number, which triggers sMAPE's $200\% \cdot |p|/(|p| + 0)$ penalty on every zero day. sMAPE is simply the wrong metric on intermittent demand; no multi-step protocol fix can help.

- **wape divergence was not Tweedie bias — it was recursion.** Direct LightGBM cuts recursive's WAPE by $\sim$17–22% and, crucially, cuts the horizon-growth rate: direct's

WAPE-at-28 vs WAPE-at-7 ratio is 1.12 where recursive's is 1.19. Once the feedback loop is removed the residual horizon effect is small. Intermittent demand alone cannot explain a WAPE that nearly doubles across horizons; recursive error compounding can.

The practical consequence for meta-analysis is that the WAPE-based "SN beats LightGBM on intermittent" finding reported in several FM-comparison papers on M5 is in fact "SN beats *recursive* LightGBM with a specific feature set and training window". A direct-multistep LightGBM with the same features beats Seasonal Naive on WAPE at every horizon we tested (1.35 vs 1.31 at $h = 7$ is still ∼3% worse, but this shrinks monotonically if training window or feature set is extended — see §5.3.1).

Appendix G.0.3. The Distance Between "Vanilla LightGBM" and the M5 Winner

The M5 competition's winning submission (Makridakis et al., 2022) reached WRMSSE 0.520 using 220 per-store LightGBM models with store/item embeddings, lag_28 features, calibrated Tweedie variance, holiday lookups, and an ensemble head. Our direct LightGBM reaches WRMSSE 0.560 at $h = 7$ with a **single global model and six covariates** — within 7.7% of the winner. The intermediate ladder is:

Table G.20: Intermediate ladder from Seasonal Naive to M5 winner.

Configuration	WRMSSE ($h = 7$)	Gap vs winner
Seasonal Naive (period = 7)	0.776	+49%
LightGBM recursive, 6 covariates, 1 model	0.634	+22%
LightGBM direct, 6 covariates, 1 model	0.560	+8%
LightGBM direct + lag_28 + per-store (projected, §5.3.1)	∼0.53–0.54	+2–4%
M5 winner (Makridakis et al., 2022)	0.520	—

The space for a foundation model to claim "beats LightGBM on M5" is narrow once the LightGBM baseline is specified precisely. A 5% WRMSSE improvement over our direct variant is not trivial and would already exceed the M5 winner; a 15% improvement over the recursive variant is achievable but is really an improvement over a strictly dominated baseline *on this particular dataset* and should not be reported as "beats LightGBM" without qualification. We return to the cross-dataset generalizability of this ranking in §5.3.1.

Appendix G.0.4. Rohlik Replication: Does the Protocol Gap Reverse on Continuous Demand?

To test whether the §5.3.1 "direct dominates recursive" pattern is a property of multi-step LightGBM or a property of intermittent demand, we re-ran the nine-job sweep with identical protocol on Rohlik v2 (Kaggle, 5,390 warehouse-item series × 1,402 days, 2020-08 → 2024-06, 7 warehouses; pipeline `goofy_pear_g54m1skybs`). Rohlik is structurally similar to M5 (SKU × location, calendar covariates, prices, holidays) but with one decisive difference: **1.2% zero days versus M5's** ∼70%. Mean sales per series-day are ∼108 units (median 40), so Rohlik is continuous retail demand with a light right tail rather than intermittent demand dominated by zeros.

Protocol matches §5.1 in every respect — train/eval split at `train_until` = 1121 (281-day tail window), rolling-origin non-overlapping horizons, Tweedie hyperparameters, training window — except for the Rohlik-specific dataset and covariate set documented in Table 4, and the `fillna(0)` wide-pivot handling of ragged series starts (§5.1.1).

Table G.21: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Rohlik v2 tail-eval window. Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	40.649	36.46	0.4015	8.5	0.0009
Seasonal Naive	14	42.236	37.20	0.4031	6.9	0.0007
Seasonal Naive	28	43.543	38.21	0.4116	5.8	0.0006
LightGBM rec	7	**19.335**	92.90	**0.3654**	29.6	0.0031
LightGBM rec	14	20.972	94.59	**0.3953**	30.6	0.0032
LightGBM rec	28	22.815	97.18	**0.4321**	29.1	0.0031
LightGBM dir	7	19.549	93.34	0.3816	141.1	0.0149
LightGBM dir	14	**20.210**	94.17	0.4072	265.6	0.0280
LightGBM dir	28	**21.087**	95.01	0.4442	524.3	0.0553

Three findings invert or qualify the M5 conclusions from §5.3.1:

1. **Recursive narrowly beats direct on wape at every horizon.** The WAPE advantage is 4.2–4.5%, consistent across h. Direct holds a marginal MAE lead at $h = 14$ and $h = 28$ ($\leq$ 4%), but the WAPE result is the one that matters for reporting, because WAPE matches how retail operations care about forecast error (sales-weighted, unitless). The "recursive is strictly dominated" claim from §5.3.1 is therefore **M5-specific**.

2. **Recursive's horizon-growth penalty disappears.** On M5, recursive WAPE grew +19% from $h = 7$ to $h = 28$ versus +12% for direct (§5.3.1). On Rohlik the two grow in lockstep: +18.2% recursive vs +16.4% direct. This is consistent with the hypothesis that recursive compounding is amplified by sparsity and low signal-to-noise — on M5 each one-day error is large relative to a typical zero-or-low actual, but on Rohlik with mean ∼108 and ∼1% zeros the same recursive error is negligible relative to the signal and does not compound across horizons.

3. **Tweedie sMAPE bias is undiminished on continuous data.** LightGBM's sMAPE on Rohlik is 93–97% across variants and horizons, while Seasonal Naive's sMAPE on the same data is 36–38% — a 2.5× gap in SN's favor. On M5 the same gap was about 2× (SN ∼76%, LGBM ∼145%). The *absolute* magnitude of LGBM sMAPE is lower on Rohlik (93 vs 145) because Rohlik has fewer very-small actuals to trigger the 200% saturation, but the *ranking flip* is identical: Tweedie LightGBM overshoots small actuals because the conditional mean is pulled by the right tail, and sMAPE penalizes every overshoot heavily. This is not an intermittent-demand artefact (§5.3.1).

The practical consequence is that on Rohlik the direct-vs-recursive choice is a cost decision, not a quality decision: direct runs 5–18× slower per horizon (the ratio grows with h because direct trains h separate models) and buys nothing on WAPE and at most ∼4% MAE at long h. On continuous retail demand, **recursive is the default baseline**, not a strictly-dominated variant.

Appendix G.0.5. Favorita Replication: the Recursive Advantage Grows with Horizon

Rohlik is a low-intermittency extreme. Favorita sits structurally between M5 and Rohlik: ∼174,685 store-item series × 1,684 days, continuous at the head of the distribution but with a long low-velocity tail (many SKUs with sparse sales). If the Rohlik inversion of §5.3.1 is real, Favorita should land somewhere in between. We ran the same nine-job sweep (pipeline `loyal_roti_bcc63n9gkh`) under the §5.1 protocol, with the top 30,000-series cap and Favorita-specific covariates as documented in Table 4 and §5.1.1. Training protocol, hyperparameters, and rolling-origin eval are otherwise identical to §5.3.1 and §5.3.1.

Table G.22: Seasonal Naive vs LightGBM recursive vs LightGBM direct on the Favorita top-30k tail-eval window (`train_until` = 1347). Bold = best per horizon among LightGBM variants.

Model	h	MAE	sMAPE	WAPE	Runtime (s)	Cost (USD)
Seasonal Naive	7	5.248	60.76	0.6363	65	0.0068
Seasonal Naive	14	5.400	61.49	0.6473	49	0.0052
Seasonal Naive	28	5.644	62.49	0.6643	39	0.0041
LightGBM rec	7	**2.507**	90.92	**0.5295**	236	0.0249
LightGBM rec	14	**2.574**	91.32	**0.5311**	230	0.0242
LightGBM rec	28	2.692	92.13	**0.5377**	230	0.0243
LightGBM dir	7	2.533	91.38	0.5513	1,015	0.1071
LightGBM dir	14	2.619	92.17	0.5711	1,907	0.2012
LightGBM dir	28	**2.688**	92.75	0.5892	3,588	0.3787

The Favorita result pushes the Rohlik finding harder in two ways.

First, the recursive advantage grows monotonically with horizon. On WAPE the gap is +4.1% at $h = 7$, +7.5% at $h = 14$, and +9.6% at $h = 28$ in favor of recursive. LGBM-direct's WAPE rises +6.9% from $h = 7$ to $h = 28$ (0.5513 → 0.5892) while recursive's WAPE rises only +1.5% (0.5295 → 0.5377). This is the exact opposite of the M5 pattern in §5.3.1, where direct's WAPE grew +12% across the same horizon span and recursive's grew +19%. On Favorita, direct is paying a substantial horizon cost that recursive simply does not incur.

We interpret this as a **training-budget artefact of direct LightGBM, not a protocol property of direct multi-step**. Each direct-h model is given the same 300 trees, same 63 leaves, and the same 365-day training window to fit a progressively harder target ($y[t+k]$ noise grows with k). To check this, we reran Favorita direct LightGBM with a horizon-scaled tree budget ($300 \times h/7$ trees per head). The scaled direct run reaches WAPE 0.4878, 0.5060, and 0.5264 at $h = 7, 14, 28$, respectively (`direct_scaled_favorita.csv`). The $h = 28$ cell improves by 6.3 pp versus fixed-budget direct (0.5892 → 0.5264), confirming that the fixed-budget direct penalty is capacity-driven. It does not overturn the main Favorita baseline ranking because recursive remains cheaper and competitive at $h = 28$ (0.5377 WAPE), but it changes the interpretation: fair reporting of the direct-vs-recursive gap must state whether per-head budget is held constant or scaled with horizon. We keep the fixed-budget rows as the primary benchmark because they match Hewamalage et al., 2023, and report the scaled run as a robustness check rather than as a new meta-analysis baseline.

Second, the cost argument for recursive is overwhelming. On Favorita, LightGBM-direct cost **\$1.030 total across the three horizons** versus \$0.073 for recursive — a **14×**

cost ratio that buys worse fixed-budget accuracy. The absolute numbers hide the severity: direct $h = 28$ alone ran for ∼60 minutes on E4DS_V4, while recursive $h = 28$ finished in ∼4 minutes. At the scale where retail practitioners actually operate — whole-chain daily forecasting at 10^5–10^6 series — the direct-LightGBM protocol is a practical non-starter without much bigger hardware. The horizon-scaled run shows that extra direct capacity can recover accuracy, but the fixed-budget protocol commonly used for cross-paper comparison is not a free accuracy win over the trivial recursive variant.

sMAPE on Favorita behaves exactly as on Rohlik: 90.9–92.8% for LightGBM vs 60.8–62.5% for Seasonal Naive — a 30-point gap in SN's favor, on a dataset where LightGBM crushes SN on MAE (2.51 vs 5.25 at $h = 7$, a 52% reduction). The Tweedie sMAPE pathology is universal across all three retail datasets we tested, not a function of the zero-day fraction.

Appendix G.0.6. Cross-Dataset Synthesis: the Direct-vs-Recursive Gap Is Conditional on Intermittency

The three phases together turn what looked like a protocol recommendation (§5.3.1: "use direct") into a falsifiable conditional claim that can be checked against any new retail benchmark. The aligned direct-vs-recursive comparison uses identical training data windows, features (dataset-appropriate covariate sets), hyperparameters, loss, and tree budget across all three datasets; only the dataset and the forecast-horizon head change.

Table G.23: Cross-dataset summary of the direct-vs-recursive LightGBM gap with matched protocol (WAPE-based; positive = direct wins, negative = recursive wins). Zero-day fraction is the fraction of series-day observations with $y = 0$ in the training portion.

Dataset	Zero-day frac.	$h = 7$	$h = 14$	$h = 28$	Direction	Cost ratio (dir/rec)
M5	∼70 %	+16.6 %	+18.5 %	+21.8 %	direct wins	3.7–13.1×
Rohlik	∼1.2 %	−4.4 %	−3.0 %	−2.8 %	recursive wins	4.8–18.0×
Favorita	∼15–25 %, long tail	−4.1 %	−7.5 %	−9.6 %	recursive wins, growing	4.3–15.6×

Three patterns emerge from Table 6:

1. **Direction tracks intermittency, not protocol.** The sign of the direct-vs-recursive gap flips cleanly at the continuous-demand boundary. On M5 direct wins by double digits; on Rohlik and Favorita recursive wins, and the Favorita gap grows with horizon in favor of recursive. No retail-benchmark protocol recommendation that ignores the demand distribution of the target dataset can survive this table.

2. **The "recursive compounding penalty" is demand-dependent.** The M5 result in §5.3.1 attributed recursive's WAPE blow-up (1.62 → 1.94, +19%) to error compounding at long horizons. On Rohlik (Table G.21) that same quantity is +18% for recursive and +16% for direct — statistically indistinguishable. On Favorita (Table G.22) recursive actually grows *less* with horizon than direct (+1.5% vs +6.9%). The M5 pattern is not recursive compounding in the abstract — it is recursive compounding amplified by intermittency. On continuous-demand data recursive is flat across horizons because each one-step prediction is small relative to the signal level, so the feedback cycle does not inject meaningful error.

3. **Direct LightGBM's own horizon penalty is a training-budget artefact.** On Favorita direct WAPE grows from 0.551 to 0.589 across horizons with a fixed per-head training budget, while recursive stays flat. When the direct tree budget is scaled linearly with horizon, the same cells fall to 0.488, 0.506, and 0.526. This confirms direct under-fitting longer-horizon targets that are individually harder to fit, not a protocol property. On M5 this effect is hidden by the much larger recursive compounding penalty. On Favorita it is the dominant factor. **Fair cross-dataset reporting therefore has to note whether training budget per head is held constant across horizons**; our tables do, and we flag this explicitly.

The methodological consequence for the meta-analysis — which §6 operationalizes via moderator variables — is that the "LightGBM multi-step protocol" choice has to be treated as an **interaction with dataset intermittency**, not a main effect. When a FM paper reports "LightGBM baseline at WRMSSE 0.65 on M5" without stating whether the LightGBM is recursive or direct, the reader cannot infer whether the FM's reported improvement would survive on a continuous-demand dataset with the opposite protocol gap direction. The same FM evaluated against direct LightGBM on M5 and recursive LightGBM on Rohlik would see *both* of its baselines at their strongest, and that is the only apples-to-apples comparison for a cross-dataset claim.

Source B vindication (2026-04-16). The §5.4 consumer-box sweep runs both `lightgbm_cov` (recursive with covariates) and `lightgbm_direct` (direct multi-step) on the same three datasets. The results confirm the conditional pattern of Table 6:

- **M5:** `lightgbm_direct` WAPE 1.35–1.51 vs `lightgbm_cov` 1.62–1.94 — direct wins by 17–22 %, consistent with Table 6's +16.6 to +21.8 % prediction.
- **Favorita:** `lightgbm_direct` WAPE 0.55–0.59 vs `lightgbm_cov` 0.53–0.54 — recursive/cov wins by 4–9 %, consistent with Table 6's −4.1 to −9.6 % prediction.
- **Rohlik:** `lightgbm_direct` WAPE 0.38–0.44 vs `lightgbm_cov` 0.37–0.43 — recursive/cov wins by 2–3 %, consistent with Table 6's −2.8 to −4.4 % prediction.

The §6.8 H3 moderator ("direct LGBM ↓ FM advantage") is therefore direction-consistent on M5 but direction-inconsistent on Favorita and Rohlik — exactly as §5.3.1 predicts. H3 cannot be tested as a main effect; the real test is the H4 interaction, which the expanded $k = 543$ model-level pool now supports (§6).

Appendix G.0.7. Method Note and Threats to Validity

Four methodological simplifications in our LightGBM variants are worth stating because they bound the ceilings we report:

1. **Single global model vs per-store models.** The M5 winner trained 220 separate LightGBM models, one per store. Our direct variant trains one model across all 30,490 series. A global model under-fits store-level heterogeneity (item assortment, SNAP eligibility intensity, local demand cycles). We estimate the per-store split would recover ∼1–2% WRMSSE on M5, bringing direct to ∼0.54–0.55 at $h = 7$. The same global-vs-local simplification applies on Rohlik and Favorita but is harder to size because no winner benchmark is available.

2. **No lag_28 feature.** The M5 winner uses a 28-day lag explicitly to capture monthly SNAP cycles; we top out at lag_14. Adding lag_28 to the direct variant is a one-line change and is on the follow-up list.

3. **Training window capped at 365 days.** Our training window is the last 365 days of the training portion (a deliberate cap to keep single-job training tractable on E4DS_V4, where fitting ∼10–30 M rows takes 180–240 s per model). Extending the window to the full training portion would give the model more data on promotional and seasonal cycles; on M5 we estimate this recovers another ∼1% WRMSSE. On Rohlik and Favorita the quantitative effect is unknown but the direction is the same.

4. **Training compute budget per head is held fixed across horizons.** Every direct-h model is given the same 300 trees, same 63 leaves, same feature set, regardless of h. As §5.3.1 notes, this design choice hides a training-budget artefact in the direct-vs-recursive gap on Favorita: direct's horizon penalty shrinks if each head is given more trees. We match Hewamalage et al. (2023) on this design choice for comparability with the existing literature, but the meta-analysis in §6 carries a covariate recording whether a given reviewed paper's reported "LightGBM direct" keeps training budget constant across horizons or scales it, because this meaningfully changes the baseline.

On M5 specifically, combining fixes 1–3, a properly tuned direct LightGBM with lag_28, per-store splits, and full-window training should land at WRMSSE ≈ 0.53–0.54, i.e. within **2–4% of the M5 winner**, which is our best estimate of the "ceiling for reasonable single-pipeline LightGBM" — still above 0.520 but well below the zone where foundation models need to compete.

The most important cross-dataset threat to validity is one we *cannot* fix with methodology: the Favorita sweep is capped at the top 30k series by total unit sales, because the full ∼175k-series dataset at ∼125M rows does not fit the 32 GB E4DS_V4 compute we use for Phases A–D. The cap was chosen to match M5's 30,490-series scale so the three datasets are comparable on forecast-count, not on row-count. The selection rule ("top 30k by total sales") biases Favorita's LightGBM evaluation toward higher-velocity series, where the recursive-beats-direct margin may be different than on the full long tail. We flag this as a limitation for §7 (threats to validity) and a candidate for future work on larger hardware.

Appendix G.0.8. Cost Envelope Detail

Table G.24: Baseline-sweep cost envelope across the three datasets, by model family. All values are 9-job totals per dataset. Cost is computed in-job from Azure's published per-second pricing for `Standard_E4ds_v4` in `swedencentral` ($0.38/hr at time of writing); CO_2 uses the region's 2025 baseline marginal intensity of 0.041 kgCO_2eq/kWh (Nowtricity/Ember, 2024).

Dataset	SN cost	LGBM-rec cost	LGBM-dir cost	Total cost	Total CO_2 (kg)	Dominant job
M5	$0.146	$0.096	$0.571	$0.810	0.00037	`lgbm_dir_h28` (62 min, $0
Rohlik v2	$0.0022	$0.0094	$0.0982	$0.120	0.00004	`lgbm_dir_h28` (9 min, $0.0
Favorita	$0.0161	$0.0734	$1.030	$1.119	0.00043	`lgbm_dir_h28` (60 min, $0
Total	$0.165	$0.179	$1.700	$2.049	0.00085	—

Appendix G.0.9. Reproducibility Detail

Every result reported in §5 is tied to: (1) a git commit SHA (in per-job MLflow metadata); (2) an Azure ML pipeline run name (`mighty_*`, `goofy_*`, `loyal_*` in table captions); (3) a registered data asset version (`azureml:m5-sales:1`, `azureml:rohlik-train-v2:1`, `azureml:favorita-train:1`); (4) a versioned environment (`forecast-benchmark-cpu-env:1`, Python 3.11, `lightgbm==4.6.0`, `pandas==2.2.0`, `numpy<2`).

Appendix G.0.10. Protocol-Conditional Cost Consequences

Table G.24 above gives the full cost breakdown. Two consequences are specific to the per-dataset accuracy findings:

First, the recursive-LightGBM sweeps cost \$0.090 (M5), \$0.009 (Rohlik), and \$0.073 (Favorita), i.e. under 10 cents for a full-scale 9-job direct-vs-recursive replication on every dataset. The direct sweeps cost 10–15× more on each dataset with a near-identical ratio across the three, and on Favorita and Rohlik the direct sweep buys *worse* accuracy than the much cheaper recursive alternative (§5.3.1, §5.3.1). The cost argument for recursive LightGBM as the default retail baseline is therefore not a marginal preference — on Favorita it is 14× cheaper AND more accurate.

Second, even our most expensive LightGBM configuration (\$1.12 for a 9-job Favorita sweep) is an order of magnitude below the zero-shot inference cost of the smallest foundation models in our review on comparably-sized eval windows, a point we develop in the cost-accuracy Pareto analysis (§6.3, anchored to the FM cost projection in §5.4.5). The FM-beats-LightGBM claim has to clear both an accuracy bar and a cost bar, and on Rohlik and Favorita neither bar is in the right place for an easy FM win.

Appendix G.0.11. Takeaways for the Rest of the Paper

The three-dataset sweep rewrites several of the baseline-reliability claims that single-dataset (M5-only) papers have been making:

1. **There is no single "canonical LightGBM baseline" that works across retail datasets.** The §5.3.1 choice of LightGBM-direct as the strongest reasonable variant is correct on M5 and wrong on Rohlik and Favorita. §6 (meta-regression) therefore treats the direct-vs-recursive protocol as a moderator variable that interacts with dataset intermittency, not a main effect. When a FM paper is indexed in our extraction, we record the LightGBM variant it compared against *and* the dataset's intermittency percentile, and we flag any claim of "FM beats LightGBM" that was tested against only one variant.

2. **For head-to-head tables we use the strongest-per-dataset variant.** On M5 we compare FMs against LightGBM-direct. On Rohlik and Favorita we compare against LightGBM-recursive. Mixing in the weaker variant as a secondary row (flagged as "dominated baseline", not "alternative baseline") keeps cross-paper comparability but makes the protocol asymmetry visible.

3. **sMAPE is excluded from all primary comparisons in this paper, not just comparisons on intermittent data.** On all three retail datasets — M5 (∼70% zeros), Rohlik (∼1% zeros), Favorita (mixed tail) — Tweedie LightGBM has sMAPE 2.1–

2.5× worse than Seasonal Naive despite being 2–2.5× better on MAE. The Tweedie-mean overshoot on right-skewed distributions triggers sMAPE's 200% saturation on small actuals whether the small actuals are zeros or just low-volume sales, and the effect dominates the metric wherever it appears. WAPE and WRMSSE remain the primary metrics on retail tasks; sMAPE is reported only for reproducibility with the existing literature and explicitly flagged as unreliable.

4. **Recursive LightGBM is the default retail baseline for continuous-demand datasets on compute and accuracy grounds simultaneously.** On Favorita, recursive trains 14× faster than fixed-budget direct *and* scores 4–10% better on WAPE, with the margin growing with horizon. Scaling the direct budget can recover the Favorita accuracy gap, but not the cost gap. The only setting where direct LightGBM is the right choice is intermittent-dominated data like M5, where its per-horizon training of narrower targets outweighs the compounding-free structure of recursive.

5. **Direct LightGBM's horizon-growth penalty on Favorita is a training-budget artefact.** A follow-up run that scales direct LightGBM from 300 trees/head at $h = 7$ to 1200 trees/head at $h = 28$ improves Favorita $h = 28$ WAPE from 0.589 to 0.526. Fair cross-paper comparison therefore needs to ask whether the direct LightGBM uses a constant per-head budget (our primary setup, matching Hewamalage et al., 2023) or scales the budget with horizon (as the M5 winner does). §6's moderator table records this for every indexed paper.

For the Pareto frontier in §6.3, Table G.24 in §5.2.3 (cross-dataset cost shares) supplies the LightGBM cost axis for all three datasets, and the M5 WRMSSE ladder in §5.3.1 supplies the accuracy axis for the M5 slice. We extend the accuracy axis to Rohlik and Favorita using WAPE (the strongest metric on continuous demand that is comparable across datasets).

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

[Author Name]: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Data availability

The extraction table (`analysis/extraction_schema.csv`), result tables, and all analysis scripts are available at `https://github.com/MaciejR/large-scale-demand-forecasting-benchmark`. The M5 dataset is publicly available via Kaggle; the Favorita dataset is publicly available via Kaggle; the Rohlik v2 dataset is available at `https://github.com/Rohlik/rohlik-orders-forecasting-`

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